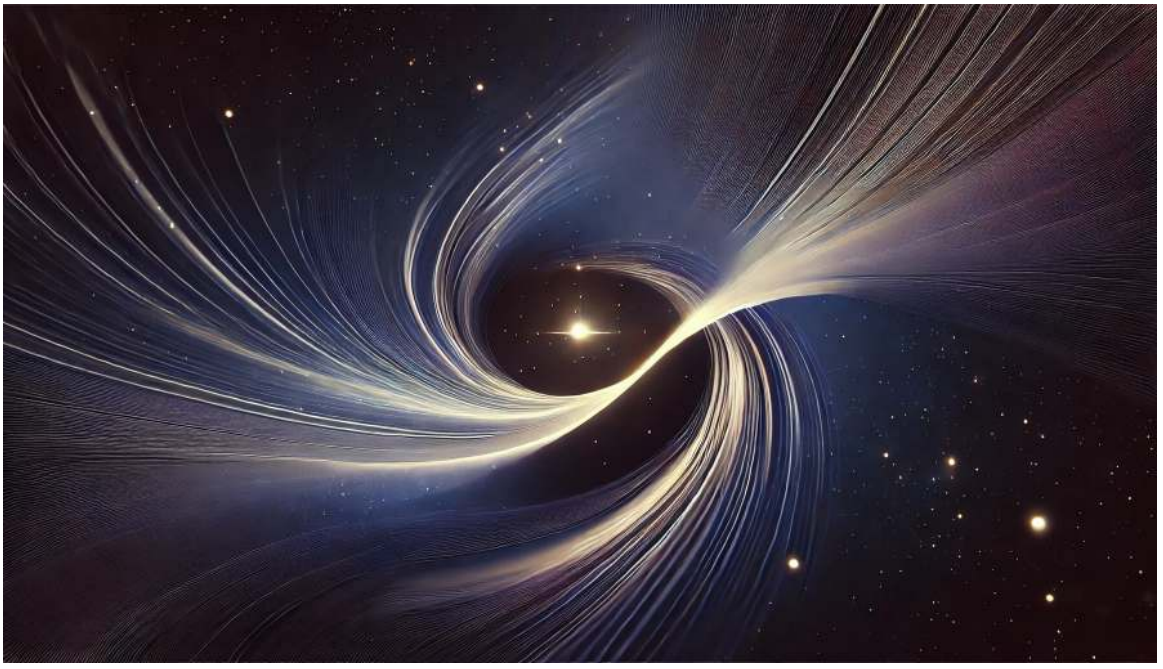


Temporal-Spatial Coupling in Gravitational Lensing: A Reinterpretation of Dark Matter Observations

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Abstract

Standard gravitational lensing analysis relies on the Isochrony Axiom—the implicit assumption that the observed image represents a synchronous spatial snapshot of the source. For evolving sources, this approximation breaks down in the presence of conformal metric couplings, creating a "temporal composite" image. This projects temporal depth onto the spatial plane, generating a temporal-composite response contribution sourced by the underlying Temporal Shear field gradient—arising from gradients in the scalar field's continuous spatial profile (Temporal Topology, TEP)—whose propagating component is confined to the millisecond (conservative-baseline) window while its chronometric component is bounded by the ratio of differential delay to source-evolution time and is accessible only through time-domain or variability-dependent observables. Within TEP, Phantom Mass is the excess dynamical mass inferred when the scalar force on matter trajectories is analyzed as gravitating mass. Its test is the mass difference inferred from dynamics and lensing at matched radii and with compatible mass models. Whereas isotropic conformal rescaling $\tilde{g}_{\mu\nu} = A^2(\phi) g_{\mu\nu}$ *stretches spatial rulers alongside clocks so* $A^2(\phi(x))$ *cancels out of null rays, under the canonical unstretched-space temporal metric* $\tilde{g}_{\mu\nu} = -A_{\rm loc}^2(u_{\rm grav}(\mathbf{x}))U_\mu U_\nu + A_{\rm sp}^2(u_{\rm drift}(t,\mathbf{x}))\Pi_{\mu\nu}$ *(*$U_\mu \equiv -\nabla_\mu\bar{\phi}/\sqrt{2\bar{X}}$*) the* $(3+1)$ *ADM spacetime lapse rescales as* $N_{\rm TEP}(x) = A(\phi(x))N_{\rm GR}(x)$ *while the spatial 3-metric* γ_{ij} *remains unstretched. Intermediate clock rates along the ray therefore contribute* $-\beta_A u_{\rm grav}(x)$ *directly to the optical refractive index* $n(x) \simeq 1 - 2\Phi_N(x)/c^2 - \beta_A u_{\rm grav}(x)$*, preserving local Lorentz invariance and* $c_{\rm GW} = c_\gamma$ *identically on the shared temporal-lapse cone (*$\Delta t_{\gamma-{\rm GW}} = 0$*). Sourced by baryons alone on the selected kinetic sector* $P_{,X} = k\sqrt{\xi} + 2\xi$ *(*$k = 16.03$ *fixed by* a_0*, zero per-system free parameters), this unstretched-space temporal refraction reproduces the Bullet Cluster galaxy-peak convergence and offset morphology (*$\kappa_{\rm tot} = 0.307$–0.372 *at the bullet galaxies, peak at* $x = +0.352$ *Mpc on the galaxies, no negative moat; Step 02) without particulate dark matter, consistent with the companion strong-lensing delay and CMB lensing closures (Papers 18, 19).*

Screening of the shear response in high-acceleration lensing environments is governed by the abstract environmental operator $\mathcal{S}_\Sigma(\mathcal{E})$*, which pins the incremental field gradient where the ambient landscape is steep while the Temporal Topology persists. The same operator sets the environment-dependence of the phantom mass, which is accordingly largest in isolated, diffuse low-acceleration systems.*

Keywords: gravitational lensing – dark matter – modified gravity – cosmology: theory – galaxies: kinematics and dynamics – temporal equivalence principle

1. Introduction

1.1 The Anomaly of the Dark Sector

The existence of dark matter, inferred from gravitational lensing (Walsh et al. 1979), cluster dynamics (Zwicky 1933), and cosmic microwave background observations (Planck Collaboration 2020), represents a significant challenge in modern physics. Despite decades of increasingly sensitive searches, no dark matter particle has been directly detected (Schumann 2019), and tensions persist between cosmological observations at different scales (Riess et al. 2022; Di Valentino et al. 2021). The prevailing paradigm assumes that these anomalies indicate the presence of an invisible *substance*. The question arises whether the apparent mass discrepancy can be resolved by relaxing the *Isochrony Axiom*—the assumption that temporal delays across an image are negligible.

Modified Newtonian Dynamics (Milgrom 1983) and its relativistic extensions (Bekenstein 2004; Skordis & Złośnik 2021) have challenged the dark matter hypothesis by modifying the gravitational force law. However, these approaches typically retain the standard metric assumptions regarding light propagation and causality. The present analysis interrogates a deeper, often unstated assumption underlying the interpretation of all astronomical signals: the nature of simultaneity in image formation.

1.2 The Isochrony Axiom

Gravitational lensing is conventionally treated as the spatial deflection of light rays by mass. Standard practice already models geometric and Shapiro time delays and, for variable sources, the light curves themselves. The closure examined here is narrower and operates one level deeper, in the reconstruction:

> **The Isochrony Axiom:** The closure in which static lens reconstructions treat the source surface-brightness distribution as effectively time-independent across the differential emission-time structure relevant to the reconstruction, after modeled propagation delays have been applied — so that source clocks, photon transport, and observer clocks are mapped onto a single general-relativistic time coordinate.

Stated this way, the axiom is not a claim that lensing analyses ignore time delays. It is the assumption that the residual emission-time structure, once known delays are removed, carries no further information — that the reconstruction may proceed as if the source were synchronous. TEP-GL names this closure and tests its breakdown.

While this axiom serves as a necessary simplification for standard analysis, it breaks down when the differential lookback time across an image becomes comparable to the evolutionary timescale of the source. Standard analysis corrects for the finite speed of light in the arrival time of signals (lookback time) but has typically neglected the differential arrival-time and clock-transfer structure across the image plane. In the presence of generalized metric couplings, this approximation fails. Gravitational lensing is fundamentally an *arrival-time* phenomenon. Images form at the stationary points of the Fermat potential, which encodes the total light-travel time (Blandford & Narayan 1986). In standard General Relativity, the difference in arrival times between images is attributed solely to geometric path differences and the Shapiro delay caused by mass. However, if the coupling between matter and gravity involves a second metric—specifically one that affects matter-clock rates or light-cone tilts—the arrival times of photons can be decoupled from the geometric definitions of "mass" used in standard GR.

If the Isochrony Axiom is violated, an observed "image" is not a snapshot of the source at one moment, but a *temporal composite*. Photons arriving at the same detector time may have left the source at significantly different emission times. For an evolving source, this temporal smearing is mathematically indistinguishable from a spatial distortion (convergence and shear) in a static reconstruction. The projected amplitude is bounded by the ratio of differential delay to source-evolution time—$\lesssim 10^{-7}$ for galaxy sources and identically zero for static sources—so the channel's domain is temporally structured sources, not the static-source lensing excess.

1.3 The Interpretive Bifurcation

Consider two mathematically equivalent interpretations of the same Fermat potential surface, distinguished only by their treatment of simultaneity:

Interpretation A (Standard Framework): Assumes the Isochrony Axiom. An Einstein ring is analyzed with apparent convergence $\kappa_{\rm obs}$ exceeding what the visible baryonic mass can produce. A dark matter halo with mass $M_{\rm DM} = M_{\rm obs} - M_{\rm baryons}$ is inferred. The "dark matter" is treated as an unseen substance required to explain the lensing geometry.

Interpretation B (TEP Framework): Rejects the Isochrony Axiom. The observed image is recognized as a *temporal composite*: photons arriving simultaneously at the detector left the source at different emission epochs, with differential emission structure set by the propagating delay surface (the gravitational Fermat surface, the unstretched-space temporal-lapse refraction, and the bounded disformal deformation) and with the conformal sector contributing the endpoint-determined clock-transfer field read by the reconstruction. For an evolving source, this temporal depth projects onto the image plane as an apparent spatial distortion, bounded in amplitude by the delay/evolution-time ratio. The differential temporal-transfer structure across the lens is computed, and apparent excess convergence in such reconstructions is identified as the signature of temporal-field gradients $\nabla(\Delta\tilde{\tau})$ in the lens environment. Where the projection applies, unmodeled time can bias reconstruction at the bounded amplitude; where it does not (stationary sources), the static lensing deflection is supplied directly by the unstretched-space temporal metric and the two-branch scalar action (§1.5, §6.9), while on the historical isotropic-conformal benchmark ($\tilde{g}_{\mu\nu} = A^2 g_{\mu\nu}$) $A^2(\phi)$ cancels from null rays.

The Critical Point: The frameworks diverge where the source is static. For evolving sources, temporal projection can contribute apparent spatial structure at the bounded amplitude and the distinction is settled by time-domain, variability-dependent, and multi-epoch observables. For a stationary backlight there is no source-evolution projection: on the historical isotropic-conformal benchmark ($\tilde{g}_{\mu\nu} = A^2 g_{\mu\nu}$) light responds only to the gravitational metric sourced by baryons and scalar stress, whereas under the canonical unstretched-space temporal metric (§1.5, §6.9) intermediate temporal depth refracts photons directly at half the dynamical scalar potential weight ($\Phi_{\rm lens} = \Phi_N + \frac{1}{2}c^2\beta_A u_{\rm grav}$ vs $\Phi_{\rm dyn} = \Phi_N + c^2\beta_A u_{\rm grav}$; Paper 0 Eq. 63), yielding both the CDM-free optical deflection and a testable dynamical–lensing mass ratio $M^{\rm u.s.}_{\rm lens}/M_{\rm dyn} = (1+\beta_A^2 y_{\rm eff})/(1+2\beta_A^2 y_{\rm eff})$.

This bifurcation illustrates that the inference of particulate dark matter is contingent, in part, on the Isochrony Axiom for temporally structured sources, while for the static-source excess it sets a sharp falsifiable boundary: where the Temporal Shear supplies the dynamical anomaly, single-pass weak lensing on isolated halos infers a smaller phantom mass than dynamics ($M^{\rm u.s.}_{\rm lens} < M_{\rm dyn}$ under the unstretched-space metric, and $g^{\rm iso}_{\rm lens} \simeq g_{\rm bar}$ on the isotropic-conformal benchmark).

1.4 The Temporal Equivalence Principle (TEP)

The Temporal Equivalence Principle (TEP), introduced in a companion paper (Smawfield 2025a), represents a shift in fundamental perspective, replacing the standard geometric framework with an operational one:

> **Temporal Equivalence Principle (TEP):** The operative physical observables in any non-local measurement are the proper-time intervals registered on physical clock worldlines, together with the phase, frequency, and arrival-time transport connecting those clock events along null signal paths. Under TEP, "gravity" includes the phenomenology of differential clock-transfer structure. The decomposition of this structure into "spatial curvature" (mass) and "temporal dilation" (metric coupling) is gauge-dependent; only the total integrated transport is invariant.

Under TEP, the central question is not "how much mass is bending the light?" but "what is the total emission-to-reception clock-transfer history associated with the signal?" In the conformal-only limit of a two-metric theory, null cones are preserved, meaning the "speed of light" is unchanged, yet the *rate* of proper time accumulation varies. This creates a disconnect between the "gravitational metric" $g_{\mu\nu}$, which carries the gravitational field dynamics and the tensor sector, and the causal "matter metric" $\tilde{g}_{\mu\nu}$, to which all nongravitational matter — test bodies, atomic clocks, and photons alike — couples universally.

The same distinction extends to cosmology. In the canonical TEP interpretation the underlying spatial scale is static, $a_m = 1$, while observed redshift is carried by the conformal clock map $A_{\rm clock} = (1+z)^{-1}$ (Athens, Paper 26; Thika, Paper 27). Quantities conventionally written as a cosmological scale factor or Hubble evolution may therefore be used as observational or reference variables without implying physical expansion of space. TEP-GL adopts this correspondence: the lensing-sector reconstruction examined here is the local multipath analogue of the cosmological temporal-to-spatial reconstruction treated in those papers.

1.5 Redefining the Dark Sector

TEP's intended ontology does not posit particulate dark matter. It accounts for both the dynamical and optical observations attributed to dark matter through the selected two-branch kinetic action $P_{,X} = k\sqrt{\xi} + 2\xi$ ($k = 16.03$ fixed by a_0, zero per-system free parameters) and the unstretched-space temporal metric $\tilde{g}_{\mu\nu} = -A^2_{\rm loc}(u_{\rm grav}(\mathbf{x}))U_\mu U_\nu + A^2_{\rm sp}(u_{\rm drift}(t,\mathbf{x}))\Pi_{\mu\nu}$ ($U_\mu \equiv -\nabla_\mu\bar{\phi}/\sqrt{2\bar{X}}$, rescaling the $(3+1)$ ADM spacetime lapse $N_{\rm TEP} = A(\phi)N_{\rm GR}$ at fixed spatial 3-metric γ_{ij}), alongside the bounded source-dependent temporal-composite response for evolving sources. It is demonstrated that:

1. **Isotropic-conformal vs. unstretched-space temporal refraction:** Under isotropic conformal rescaling $\tilde{g}_{\mu\nu} = A^2(\phi)g_{\mu\nu}$, spatial rulers stretch by $A^2(\phi)$ alongside clocks so $A^2(\phi(x))$ cancels out of $d\tilde{s}^2 = 0$, leaving only the Planck-suppressed scalar stress tensor ($\kappa_\phi \simeq 6.5\times10^{-6}$ on the selected two-branch sector; Paper 19, Step 64) and the bounded spatial-disformal delay (Steps 61–63). Under the unstretched-space temporal metric $\tilde{g}_{\mu\nu} = -A^2_{\rm loc}(u_{\rm grav}(\mathbf{x}))U_\mu U_\nu + A^2_{\rm sp}(u_{\rm drift}(t,\mathbf{x}))\Pi_{\mu\nu}$ ($U_\mu \equiv -\nabla_\mu\bar{\phi}/\sqrt{2\bar{X}}$), space does not stretch and every intermediate point x along the photon path contributes its local

temporal depth $-\beta_A u_{\rm grav}(x)$ directly to the optical refractive index $n(x) \simeq 1 - 2\Phi_N(x)/c^2 - \beta_A u_{\rm grav}(x)$, reproducing the Bullet Cluster galaxy-peak convergence ($\kappa_{\rm tot} = 0.307$–0.372, Step 02), the SN Refsdal delay and arrival order (Paper 19, Step 64), and the CMB lensing cosmic-web boost (Paper 18, Step 21) from baryons and the selected two-branch action $P_{,X} = k\sqrt{\xi} + 2\xi$ without particulate dark matter.

2. **GW170817 is a differential constraint:** The multi-messenger constraint $|c_\gamma - c_g|/c \lesssim 10^{-15}$ is explicitly reanalyzed. It is shown that this bounds only the *disformal* (cone-tilt) component of the coupling. Both isotropic conformal rescaling and the unstretched-space temporal metric preserve $c_{\rm GW} = c_\gamma$ identically because transverse-traceless tensor perturbations of γ_{ij} and electromagnetic waves evolve in the same physical ADM lapse $d\tilde{t} = A(\phi)\, dt$ on the shared unstretched spatial geometry (evading the $\sim 10^7$ s halo arrival-time split of photon-only disformal dark-matter emulators; Boran et al. 2018), while quadratic-branch kinetic screening $y \sim (r/r_*)^{4/3} \ll 10^{-5}$ satisfies Solar-System PPN Shapiro bounds.
3. **Regime I vs. Regime II:** The conservative baseline (Regime I) is the disformal multipath deformation at its GW170817-bounded normalization ($\Delta\tau_B = \tfrac{1}{2}\int D(\partial_{\hat{n}}\varphi)^2\, dl \lesssim 0.2$ s per 2 Mpc), the even-parity channel of the same $B(\varphi)$ sector the holonomy program bounds on the odd channel — so that the stated <0.1 ms lensed-FRB null threshold operates as a bound on the disformal normalization itself. Regime II collects the reconstruction-space and time-domain content of the conformal sector — endpoint clock calibration, chronometric bookkeeping, and the source-dependent temporal-composite response — whose image-space amplitude is bounded by the delay/evolution-time ratio, alongside the source-independent unstretched-space temporal refraction evaluated in Section 6.9 (Step 02).

Relaxing the isochronous reconstruction identifies bounded, testable temporal contributions to mass inference alongside the source-independent optical refraction of the unstretched-space temporal metric across galaxy, cluster and CMB lensing.

TEP does not posit an alternative dark substance. Its replacement of particulate dark matter uses the same selected two-branch action $P_{,X} = k\sqrt{\xi} + 2\xi$ and baryon distribution to explain both the dynamical and optical channels: Temporal Shear bends slow matter trajectories while intermediate temporal depth $-\beta_A u_{\rm grav}(x)$ across unstretched space refracts photons along the line of sight.

This paper does not deny any of the observations attributed to dark matter. It distinguishes real lensing, dynamical extra-force inference and source-dependent temporal reconstruction so that none is counted as an explanation of the others without an observable-level transfer calculation.

The claim-discipline framework for the TEP corpus, including the scope limitations of canonical precision tests, is established in TEP-EXP (Paper 9).

2. Theoretical Framework

2.1 The Two-Metric Postulate

The gravitational field dynamics and the matter sector are posited to be governed by distinct metrics related by a scalar field ϕ. The *Gravitational Metric* $g_{\mu\nu}$ carries the gravitational field dynamics and the tensor sector: it has Einstein–Hilbert form and gravitational waves propagate on its null cones. All nongravitational matter — including massive test bodies, atomic clocks, and electromagnetic fields — couples universally to the *causal Matter Metric* $\tilde{g}_{\mu\nu}$, on which nongravitational dynamics, signals, and quantum phases evolve. This universal coupling is the foundational TEP structure (Jakarta, Paper 0); environmental screening can render matter-frame trajectories consistent with the GR limit to high precision. The general two-metric relation (Bekenstein 1993) is adopted:

$$\tilde{g}_{\mu\nu} = A^2(\phi) g_{\mu\nu} + B(\phi) \nabla_\mu \phi \nabla_\nu \phi$$

Here, $\tilde{g}_{\mu\nu}$ is the metric on which all nongravitational matter propagates, and $g_{\mu\nu}$ is the metric carrying the gravitational field dynamics. The function $A(\phi)$ defines the *Conformal Sector* (isotropic scaling of proper time/length) and $B(\phi)$ defines the *Disformal Sector* (anisotropic stretching along field gradients). TEP introduces the temporal field as the additional physical degree of freedom: all physical rulers and clocks are built from matter coupled to $\tilde{g}_{\mu\nu}$, while the temporal field's stress-energy and strong-field operators backreact on the Einstein-frame geometry $g_{\mu\nu}$. The structure is therefore not a reinterpretation of measurement alone; sufficiently strong temporal structure induces genuine geometric backreaction (Bahrain, Paper 28), and on the historical isotropic-conformal benchmark ($\tilde{g}_{\mu\nu} = A^2(\phi) g_{\mu\nu} + B(\phi)\nabla_\mu\phi\nabla_\nu\phi$) it is that backreaction — together with the disformal sector — which carries any modification of null-trajectory lensing, whereas under the canonical unified unstretched-space temporal metric $\tilde{g}_{\mu\nu} = -A_{\rm loc}^2(u_{\rm grav}(\mathbf{x}))U_\mu U_\nu + A_{\rm sp}^2(u_{\rm drift}(t,\mathbf{x}))\Pi_{\mu\nu} + B(\phi)\Pi^\alpha_\mu \Pi^\beta_\nu \nabla_\alpha\phi\nabla_\beta\phi$ (Paper 0 §2.2, Step 80; evaluated in §1.5 and §6.9 below) intermediate temporal depth $A_{\rm loc}(u_{\rm grav})$ along the cosmic temporal flow refracts photons directly across the unstretched spatial manifold.

Box 2.0: Sector Dictionary — Three Projections of One Temporal Field

TEP-GL distinguishes three observational projections of the same temporal field. They are complementary and must not be collapsed into a single optical effect. This distinction organizes the remainder of the paper.

1. **Conformal open-path transport** $A(\phi)$: changes the matter-clock transfer associated with geometrically distinct source–observer paths. On the isotropic-conformal benchmark ($\tilde{g}_{\mu\nu} = A^2 g_{\mu\nu}$), this alters arrival-time, distance, velocity, and mass *reconstructions* without changing the unparameterized null trajectories, whereas under the canonical unstretched-space metric ($\tilde{\gamma}_{ij} = \gamma_{ij}$, §6.9) intermediate temporal depth $A_{\rm loc}(u_{\rm grav})$ also enters the optical refractive index directly.
2. **Geometric optical response** $g_{\mu\nu}[\phi]$, $B(\phi)$, and unstretched-space $A_{\rm loc}(u_{\rm grav})$: scalar backreaction on the Einstein-frame metric, any active disformal contribution, and — under the canonical unstretched-space temporal metric (§6.9, Step 02) — intermediate temporal lapse depth change the optical tidal matrix and therefore produce genuine angular convergence and shear.
3. **Temporal-composite response:** for an evolving or moving source, a path-dependent emission-time map produces an additional source-dependent image distortion.

Accordingly, *Phantom Mass* is the apparent non-particulate dark component generated by the combined temporal-field geometry and its reconstruction under the Isochrony Axiom:

$$M^{\rm TEP}_{\rm phantom} = \underbrace{M^{A}_{\rm rec}}_{\text{chronometric reconstruction}} + \underbrace{M^{g[\phi],B}_{\rm opt}}_{\text{coherent optical}} + \underbrace{M_{\rm TC}}_{\text{source-dependent}} .$$

These are not three unrelated mechanisms substituted for particulate dark matter. They are three observational projections of one dynamical temporal field. Standard analysis combines them into a single inferred mass distribution precisely because it assumes a universal isochronous mapping between source clocks, photon transport, and observer clocks. The three terms also sit on different carriers, which the remainder of the paper keeps explicit: $M^{g[\phi],B}_{\rm opt}$ is a propagated optical contribution (backreacted geometry, the bounded disformal deformation, and under the canonical unstretched-space metric of §6.9 the direct refractive response to $A_{\rm loc}(u_{\rm grav})$); $M_{\rm TC}$ propagates only through the same carriers — the isotropic-conformal contribution to a propagated emission-time map is identically zero; and $M^{A}_{\rm rec}$ lives entirely in reconstruction space, entering through the endpoint-determined clock-transfer field and the inference degeneracy (mass-sheet and normalization class) rather than through any propagated delay on the isotropic-conformal benchmark.

Box 2.1: Relation to Established Scalar-Tensor Theories

TEP is a specific two-metric scalar–tensor effective field theory, distinguished by the universal coupling of all nongravitational matter to a single causal metric and by its dynamical-proper-time interpretation. Its covariant operators draw on established scalar–tensor constructions, while its operational ontology and observable sector decomposition are specific to TEP. The correspondences are:

- **Brans-Dicke Theory:** With $B = 0$, $A(\phi) = e^{\beta_A \phi / M_{\rm Pl}}$, and the corresponding kinetic and potential choice, the conformal sector contains a Brans–Dicke-like limit in the Jordan frame, with β_A related to the Brans–Dicke parameter ω_{BD}. The correspondence is a limit of the conformal sector, not an identity of the full theory.
- **Horndeski and Beyond:** Higher-derivative scalar-tensor theories (Horndeski, DHOST) provide candidate microscopic completions for the gradient-dependent branch of $\mathcal{S}_\Sigma(\mathcal{E})$. TEP-GL does not commit to a specific completion; the canonical operator is defined at the theory level independently of any particular Lagrangian.
- **Screening and Temporal Topology:** The core TEP framework formulates screening through the environmental operator $\mathcal{S}_\Sigma(\mathcal{E})$ acting on the Temporal Shear Σ_μ, a continuous spatial profile. Imported modified-gravity screening mechanisms are not part of the TEP ontology; the screening object is $\mathcal{S}_\Sigma(\mathcal{E})$ alone.
- **DHOST Theories:** Degenerate Higher-Order Scalar-Tensor theories extend Horndeski by imposing a degeneracy condition that eliminates the additional propagating mode, rather than by keeping the field equations manifestly second-order. TEP is compatible with this broader class but does not require it.

Key Distinction: TEP's distinguishing content is the universal causal-metric coupling together with the *observational interpretation* it licenses — specifically, the recognition that conformal coupling creates a "temporal composite" image whose residual timing structure a standard reconstruction absorbs as mass. Individual covariant operators are drawn from established scalar–tensor constructions; the sector dictionary, the universal coupling, and the resulting lensing phenomenology are specific to TEP.

2.2 The Chronometric Lensing Framework

TEP-GL lensing contains three projections of one temporal field: coherent optical response through scalar backreaction and any permitted disformal contribution; conformal open-path chronometric reconstruction; and the source-dependent temporal-composite response.

1. The Static Clock-Transfer Contribution (Chronometric Reconstruction)

The conformal clock sector modifies the relation between accumulated matter-frame time and the geometry reconstructed under an isochronous model. The associated effective clock-transfer contrast is:

$$\Delta\tilde{\tau}_{\rm static} = \frac{1}{c}\int (A(\phi) - 1)\, dl$$

In the phenomenological isochronous reconstruction used here, this is represented as a source-independent correction to the inferred arrival-time surface. In multipath configurations, it is therefore operationally degenerate with the mass normalization inferred from multipath timing, distance, and joint image–delay reconstruction. The angular positions of Einstein rings and major arcs remain governed by the optical geometry.

The expression above is a matter-clock transfer functional evaluated on a timelike clock congruence associated with the path; it is not a photon observable, and for photons its differential content vanishes identically. That vanishing is a statement about a restricted net observable — the static conformal contribution to an inter-image arrival-time contrast — not that the photon is unaffected while traversing the topology: the locally measured frequency rises descending into each well and falls climbing out, asymmetric crossings slice residuals to the intervening field difference, and lapse-geometry dwell through each structure accrues monotonically. Only the frequency bookkeeping of a strictly static conformal field telescopes to the endpoints. Two equivalent statements fix this on the isotropic-conformal benchmark ($\tilde{g}_{\mu\nu} = A^2(\phi) g_{\mu\nu}$, where spatial rulers stretch alongside clocks). First, under isotropic conformal rescaling, null cones are conformally invariant (Axiom 1, Appendix A.1): $A(\phi)$ drops out of the null transport equation, so the coordinate travel time along any fixed spatial path is identical in $\tilde{g}_{\mu\nu}$ and $g_{\mu\nu}$, and the only conformal content in an observed arrival time is the endpoint clock ratio — a common-mode factor for all images of one system. Second, the underlying Temporal Shear is an exact one-form, $\Sigma_\mu = \nabla_\mu \ln A$: its open-path integral is endpoint-determined, so the shear-sector contribution to any path contrast between rays joining the same source and observer cancels algebraically, before screening is even applied. The dedicated strong-lensing transport pipeline confirms the result directly on the isotropic-conformal benchmark: the exact additional static-conformal inter-image residual is 0.0 d (TEP-LENS, Paper 19, step_56). The functional $\Delta\tilde{\tau}_{\rm static}$ is therefore retained in a sharply delimited role: it is the chronometric reconstruction map — the field of clock-rate contrasts that an isochronous lens model would misassign to mass-

sheet normalization, distance, or time-delay calibration if it were read as propagating emission-time structure — and it bounds, rather than supplies, the isotropic-conformal arrival-time residual. Genuine path-dependent null delay and deflection beyond the GR geometry arise when space does not stretch with $A(\phi)$ — specifically, under the canonical unified unstretched-space temporal metric $\tilde{g}_{\mu\nu} = -A_{\rm loc}^2(u_{\rm grav}(\mathbf{x}))U_\mu U_\nu + A_{\rm sp}^2(u_{\rm drift}(t,\mathbf{x}))\Pi_{\mu\nu} + B(\phi)\Pi_\mu^\alpha\Pi_\nu^\beta\nabla_\alpha\phi\nabla_\beta\phi$ ($U_\mu \equiv -\nabla_\mu\bar{\phi}/\sqrt{2\bar{X}}$; Paper 0 §2.2, Step 80, evaluated in §6.9 below), where the $(3+1)$ ADM lapse $N_{\rm TEP} = A_{\rm loc}(u_{\rm grav})N_{\rm GR}$ rescales along the cosmic temporal flow while the spatial 3-metric $\tilde{\gamma}_{ij} = \gamma_{ij}$ remains unstretched at fixed epoch so intermediate temporal depth $-\beta_A u_{\rm grav}(x)$ refracts null rays directly via $n(x) \simeq 1 - 2\Phi_N/c^2 - \beta_A u_{\rm grav}(x)$ — or through gravitational backreaction in $g_{\mu\nu}$ and the spatial-disformal sector $B(\phi)$: pure isotropic multiplication by $A^2(\phi)$ does not bend a photon onto a new null trajectory and does not shift its arrival time relative to any other photon on the same endpoints. Under the isotropic-conformal benchmark the lensing signal is a reconstruction-space discrepancy, whereas under the canonical unstretched-space temporal metric (§6.9, Step 02) intermediate temporal depth supplies direct optical refraction alongside the chronometric reconstruction map.

2. The Source-Evolution Response (Temporal Lensing)

While the static term reshapes the clock-transfer field read by an isochronous reconstruction, the gradient of the differential delay field $\nabla(\Delta\tilde{\tau})$ acts as a "shutter" that modulates the arrival time of photons from different parts of the source. For a source with evolution or motion, this creates a *temporal-composite response*:

$$\gamma^{\rm TC} \propto \left(\frac{\partial I}{\partial t} + \vec{v}_s \cdot \nabla I\right)\nabla(\Delta\tilde{\tau})$$

The nomenclature follows the canonical corpus convention: *Temporal Shear* denotes the field gradient $\Sigma_\mu \equiv \nabla_\mu \ln A(\phi)$, which generates the path-dependent clock-transfer field, whereas *temporal-composite response* $\gamma^{\rm TC}$ denotes the observable source-dependent image response to that field. The two must not be conflated: the former is a property of the temporal field, the latter of the observation.

This source-evolution response supplies a source-dependent phantom-mass component — an apparent mass that can fluctuate with source type. It operates as a re-ordering of emission epochs rather than a deflection of rays, and for galaxy sources it is bounded by the ratio of differential delay to source-evolution time at $\lesssim 10^{-7}$ in shear, far below the observed lensing excess. (See Section 3 for the full derivation).

Together, these two mechanisms constitute the TEP framework: the clock-transfer contribution may contribute to the phenomenology conventionally attributed to the dark matter halo, and the source-evolution response may account for part of the apparent "complexity" of substructure. Any genuinely new null-trajectory bending beyond the conformally related geometry must arise through gravitational backreaction in $g_{\mu\nu}$, a non-negligible disformal contribution, or another explicitly derived part of the coupled field solution.

2.3 Operational Axioms: The TEP Framework

The TEP framework rests on four foundational axioms. These are not approximations or perturbative corrections to General Relativity; they constitute a complete operational framework for interpreting gravitational phenomenology. They are adopted as *primary postulates* from which observational consequences are derived:

Axiom 1 (Causal Universality): In the Conformal Limit ($B = 0$), the null cones of $g_{\mu\nu}$ and $\tilde{g}_{\mu\nu}$ are identical. Photons and gravitational waves follow the same null geodesics. No "speed of light" difference exists in this limit. *This axiom is exact, not approximate.*

Axiom 2 (Proper Time Primacy): The fundamental observables in any timing measurement are the proper-time intervals $\Delta\tilde{\tau}$ registered on physical emitter and receiver clock worldlines, together with the phase, frequency, and arrival-time transport connecting those clock events along $\tilde{g}$-null signal paths. Coordinate time differences Δt are not observables; they are inferred quantities dependent on the metric model. Null proper time along the photon trajectory is identically zero (Jakarta); wherever this paper refers to accumulated path time, the phrase denotes the operational clock-transfer functional associated with the null path, not proper time experienced by the photon. *This axiom establishes clock-registered proper time as the irreducible physical observable; all other timing quantities are derived.*

Axiom 3 (Non-local temporal transport): TEP admits two distinct classes of non-local timing observable.

(a) Synchronization holonomy: A genuinely closed, direction-reversing transport loop can exhibit residual non-closure only when the transport connection contains non-exact structure, such as the disformal $B(\phi)$ sector. In the pure conformal limit, the $A(\phi)$ contribution is exact and its residual closed-loop integral vanishes.

(b) Differential path transport: Distinct open propagation paths may accumulate different temporal-transfer corrections. In gravitational lensing, two images correspond to two open paths γ_i, γ_j, and the observable is the differential transport residual

$$\Delta\mathcal{T}_{ij}^{\rm resid} = [\mathcal{T}[\gamma_i] - \mathcal{T}[\gamma_j]] - [\mathcal{T}_{\rm GR}[\gamma_i] - \mathcal{T}_{\rm GR}[\gamma_j]],$$

where $\mathcal{T}[\gamma] \equiv \Delta\tilde{\tau}[\gamma]$ is the time-transport functional. This is an *open-path blind-prediction residual*, not a closed-loop synchronization holonomy. Two structural properties delimit it. First, because each image has a single observed arrival time, algebraic closure of measured pairwise delays around image triplets vanishes identically: the axiom alters the relation between lens geometry and observed arrival time without violating the algebraic identity among the observed arrival times. Second, in the pure conformal sector the residual itself vanishes identically, because $\Sigma_\mu = \nabla_\mu \ln A$ is an exact one-form and its open-path integrals are endpoint-determined (§3.1.3); a nonzero $\Delta\mathcal{T}_{ij}^{\rm resid}$ is therefore a signature of non-exact transport — the disformal sector or scalar backreaction — not of the conformal clock-transfer map.

This axiom distinguishes open-path differential residuals (the GL observable) from closed-loop holonomy (the domain of triangle time-transfer and direction-reversing experiments), and is consistent with the refined strong-lensing formulation in TEP-LENS (Paper 19).

Measurement Protocol for the GL Observable. The lensing-sector observable is not algebraic delay closure. It is a blind-prediction residual. For each image pair (i, j), compare the observed delay $\Delta t_{ij}^{\rm obs}$ with the pre-specified GR lens-model prediction $\Delta t_{ij}^{\rm GR}$:

$$R_{ij} = \Delta t_{ij}^{\rm obs} - \Delta t_{ij}^{\rm GR}.$$

TEP predicts that these residuals should correlate with temporal-transport tracers such as projected potential depth, magnification, variability timescale, or other lens-environment proxies. Algebraic closure of observed pairwise delays is not a TEP discriminator, because it vanishes identically whenever each image has a unique arrival time.

Systematic Error Budget: Several astrophysical systematics can produce apparent residuals that must be distinguished from true temporal-transport effects:

- **Lens Model Degeneracies:** The mass-sheet degeneracy and source-position transformation can bias predicted delays. These affect the *predicted* delay (from the model), not the *observed* delay. The test compares observed delays against model predictions, and systematic lens-model error must be propagated into the residual uncertainty.
- **External Convergence:** Line-of-sight structure acts approximately as an external mass-sheet transformation, rescaling the Fermat-potential and time-delay normalization. If omitted, it can generate coherent biases in R_{ij}. Each lens must therefore marginalize over $\kappa_{\rm ext}$ and the associated mass-sheet uncertainty before testing for a TEP residual.
- **Microlensing:** Stellar microlensing in the lens galaxy can shift apparent arrival times by hours to days. However, microlensing is stochastic and uncorrelated between images; over an ensemble of lens systems, microlensing-induced residuals should average to zero with RMS scaling as $1/\sqrt{N}$.
- **Host Galaxy Delays:** Differential extinction or scattering in the host can introduce chromatic delays, but TEP's temporal-transport residual is achromatic. Chromatic residual patterns indicate astrophysical contamination, not metric effects.

Discriminator: True temporal-transport residuals should produce a *systematic* non-zero regression coefficient or covariance between R_{ij} and a pre-specified temporal-transport predictor, using a fixed image-pair ordering and hierarchical marginalization over lens-model degeneracy, $\kappa_{\rm ext}$, microlensing, and host-galaxy delays, while astrophysical systematics produce *random* residuals that average to zero across multiple systems. An unconditional mean could cancel even when a real path-dependent correlation exists. The correlation is the correct TEP discriminator.

Axiom 4 (Screening and Temporal Topology): Screening manifests as a continuous spatial profile (Temporal Topology) governed by the non-linear superposition of field gradients (Temporal Shear), suppressing the Temporal Shear where the ambient time-gradient is steep (weak-field: high-acceleration environments) while leaving cosmology accessible to dynamics. The suppression of local Temporal Shear in deep potential environments continuously reduces

$$\Sigma_\mu^{\rm obs} = \mathcal{S}_\Sigma(\mathcal{E})\,\Sigma_\mu.$$

Channel-specific PPN responses are evaluated only after this environmental projection is applied. Existing multi-messenger constraints (e.g., GW170817) are interpreted within the standard framework that assumes a single metric governs all sectors; in the TEP framework, these constraints apply to specific parameter combinations (primarily the disformal sector) and do not constitute blanket exclusions of two-metric effects. *This axiom establishes that environmental suppression is a geometric property of the field's spatial profile, not a binary boundary condition, and that constraints derived under Isochrony-assuming frameworks must be re-derived operationally within TEP before being applied as exclusions.*

These axioms are mutually consistent and together define the TEP interpretation of gravitational phenomenology. They replace the implicit Isochrony Axiom of standard lensing analysis with an explicit dynamical-time framework.

The Hidden Closure in Standard Practice: In addition to the Isochrony Axiom, most observational inference quietly assumes that time transport is globally integrable: after correcting for known effects (Sagnac, Shapiro, troposphere, etc.), a single global time coordinate can be assigned such that closed-loop synchronization holonomy vanishes. Operationally, this is the step that licenses the non-local conversion $d = c\,t$ and the inference "timing residual $\Rightarrow$ mass residual." In TEP, this closure is not assumed; Axiom 3 treats open-path differential residuals as the primary observables, while true synchronization holonomy requires direction-reversing closed loops or non-exact transport structure.

2.4 Conformal vs. Disformal Phenomenology

The distinction between the two sectors is critical for interpreting multi-messenger constraints, and it rests on a fundamental distinction between *Single-Path* and *Multipath* measurements. Furthermore, it necessitates a redefinition of the "speed of light":

- **Conformal Sector ($A(\phi)$):**
  - **Geometry:** Preserves angles and null cones. $\tilde{g}_{\mu\nu}k^\mu k^\nu = A^2(\phi)g_{\mu\nu}k^\mu k^\nu = 0$.
  - **Local Invariance vs. Global Variability:** Local c remains invariant (measured as $299,792,458$ m/s by any local clock; Paper 0, Theorem 2). What varies globally is not any speed of propagation, but the *inferred ratio between spatial separation and matter-clock transfer registered between endpoints* along extended paths. Because the rate of proper time accumulation $d\tilde{\tau} = A(\phi)d\tau_g$ varies with location, the clock transfer associated with traversing a fixed spatial interval depends on the scalar field value: a timelike clock transported through a halo well ($A < 1$) accumulates less matter-frame time than the ambient rate, and one transported through a void ($A > 1$ relative to ambient) accumulates more. For photons this path-integrated content is not observable at all — the null condition is conformally invariant — and the conformal sector enters the photon sector only through the endpoint clock ratio, the same channel that carries cosmological redshift (Paper 0, §2.2). Any "variable speed of light" language is therefore shorthand for an operational endpoint ratio inferred under a universal-clock assumption, never a change in local propagation.
  - **Single-Path Physics (GW170817):** Photons and gravitational waves from the exact same source coordinate follow the *same* null geodesic. Any temporal distortion $A(\phi)$ along this path is common-mode. The signals do not diverge because they share the same history.
  - **Multipath Physics (Lensing):** Gravitational lensing involves light rays taking *different* paths around a mass distribution. These paths traverse different regions of the scalar field $\phi(\vec{x})$, but photon transport along them is conformally invariant: the conformal differential clock-transfer contrast is a reconstruction-space field, not a propagated delay, and it generates the chronometric component of the Phantom-Mass signature only as an inference degeneracy. Genuine propagated differential delay is carried by scalar backreaction (absorbed by lens models as mass) and by the GW170817-bounded disformal deformation; coherent optical convergence and shear arise through the same two channels.
- **Disformal Sector ($B(\phi)$):**
  - **Geometry:** Tilts null cones. The effective speed of light differs from the speed of gravity: $c_\gamma \neq c_g$.
  - **Observables:** Differential arrival times between species, constrained by GW170817 to $|c_\gamma - c_g|/c \lesssim 10^{-15}$ for the path-averaged monopole.

2.5 The "Phantom Mass" Mechanism

Standard lensing reconstruction solves for a mass distribution $\Sigma(\vec{\theta})$ that reproduces the observed image distortions. This reconstruction assumes the Isochrony Axiom: $I_{\rm obs}(\vec{\theta}) = I_{\rm src}(\vec{\beta})$. However, in the TEP framework, the observed image at a given observation time $t_{\rm obs}$ is the source evaluated at a shifted effective time:

$$I_{\rm obs}(\vec{\theta},\, t_{\rm obs}) = I_{\rm src}\Big[\vec{\beta}_{\rm opt}(\vec{\theta}),\; t_{\rm obs} - \Delta T_{\rm eff}(\vec{\theta})\Big],$$

where $\Delta T_{\rm eff}(\vec{\theta})$ is the effective emission-time map along the line of sight at image position $\vec{\theta}$, and $\vec{\beta}_{\rm opt}(\vec{\theta})$ is the optical ray map generated by the scalar-backreacted and disformal geometry. The propagating content of $\Delta T_{\rm eff}$ decomposes into the ordinary geometric-plus-Shapiro Fermat surface already contained in any GR lens model, the disformal multipath deformation bounded by GW170817 ($\lesssim 0.2$ s per 2 Mpc halo path; Box 3.2), and the common-mode endpoint clock ratio; the conformal sector contributes no path-dependent term, because null transport is conformally invariant (§3.1.1) and its clock-transfer map is endpoint-determined (§2.2). For a finite exposure over $[t_1, t_2]$, the recorded image is the time-averaged quantity:

$$\bar{I}_{\rm obs}(\vec{\theta}) = \frac{1}{\Delta t_{\rm exp}} \int_{t_1}^{t_2} I_{\rm src}\Big[\vec{\beta}_{\rm opt}(\vec{\theta}),\; t - \Delta T_{\rm eff}(\vec{\theta})\Big]\, dt.$$

For a spatially varying temporal field, $\Delta T_{\rm eff}$ varies across the image plane through its propagating components — the gravitational Fermat surface and the disformal deformation — while its conformal clock-transfer content varies only in reconstruction space. If the source has temporal variability (secular evolution, rotation, or fluctuations) on the timescale of $\nabla_\theta(\Delta T_{\rm eff})$, the finite-exposure average smears the recorded image.

The Equivalence: A gradient in the emission-time map across an image is mathematically equivalent to a shearing of the source frame. To a static observer assuming isochrony, this "temporal shear" is indistinguishable from the "gravitational shear" caused by mass. Thus, for temporally structured sources, *purely temporal structure is misinterpreted as "Phantom Mass" (dark matter).* The channel is bounded at $\lesssim 10^{-7}$ for galaxy sources and vanishes for static sources; it is a source-dependent discriminator, not the carrier of the observed static-source lensing excess.

One Quantity, Two Measurements

Phantom Mass is not a separate mechanism in each observable channel; it is one quantity — the mass a standard analysis infers because it does not model the TEP temporal field. It enters through two routes: in dynamics, where the Temporal Shear supplies extra apparent acceleration ($M_{\rm ph}^{\rm dyn}$); and in reconstruction, where the differential source-clock history of an evolving source is read as spatial structure ($M_{\rm ph}^{\rm recon}$, zero for static sources). Each channel is inferred independently: $M_{\rm ph}^{\rm dyn} \equiv M_{\rm dyn}^{\rm GR} - M_{\rm baryon}$ and $M_{\rm ph}^{\rm lens} \equiv M_{\rm lens}^{\rm GR} - M_{\rm baryon}$. Comparing them is then a *test* of Phantom Mass rather than its definition — the empirical question is $M_{\rm ph}^{\rm dyn} \overset{?}{=} M_{\rm ph}^{\rm lens}$. For a single shared collisionless mass distribution, matched dynamical and lensing analyses should recover mutually consistent mass profiles within orbital-anisotropy, projection, baryonic-modelling, and external-convergence uncertainties. In TEP, on the historical isotropic-conformal benchmark ($\tilde{g}_{\mu\nu} = A^2 g_{\mu\nu}$) the Temporal Shear acts on matter trajectories while $A^2(\phi)$ cancels from photon paths ($M_{\rm ph}^{\rm lens,iso} \simeq 0$), whereas under the canonical unstretched-space temporal metric ($\tilde{\gamma}_{ij} = \gamma_{ij}$, §1.5 and §6.9) intermediate temporal depth refracts photons at half the dynamical scalar potential weight ($\Phi_{\rm lens} = \Phi_N + \frac{1}{2}c^2\beta_A u_{\rm grav}$ vs $\Phi_{\rm dyn} = \Phi_N + c^2\beta_A u_{\rm grav}$; Paper 0 Eq. 63, $M_{\rm lens}^{\rm u.s.}/M_{\rm dyn} = (1+\beta_A^2 y_{\rm eff})/(1+2\beta_A^2 y_{\rm eff})$). In both realizations the single-halo prediction is $M_{\rm ph}^{\rm dyn} > M_{\rm ph}^{\rm lens}$, with the difference largest where the shear is active — the isolated, low-acceleration systems where $\mathcal{S}_\Sigma \to 1$ (approaching parity $M_{\rm lens}^{\rm u.s.} \simeq M_{\rm dyn}$ on multi-component or group-embedded sightlines). This phantom-mass consistency test is the framework's decisive dark-sector check: agreement would bound the single-halo differential between dynamical shear and optical temporal refraction; a systematic positive difference at matched radii, persisting after anisotropy, projection, and mass-sheet systematics are marginalized, is a distinctive TEP signature not generically expected from a single shared collisionless mass distribution.

2.6 Two Regimes of TEP-GL

Box 1: The Two Observational Regimes

The TEP-GL framework operates in two distinct regimes, distinguished by the magnitude of the differential clock-transfer residual $\Delta\tilde{\tau}$ across the lens:

Regime I: the conservative baseline

- **Physical carrier:** The disformal multipath deformation. For photons on the matter metric the phase velocity is $v_{\rm ph}/c \simeq 1 - \frac{1}{2}D(\partial_{\hat{n}}\phi)^2$ with $D \equiv B/A^2$ (Paper 0, App. B); a differently routed image pair traverses different $\nabla\phi$ structure, so the inter-image residual is $\Delta\tau_B = \frac{1}{2}\int D(\partial_{\hat{n}}\phi)^2\, dl$, evaluated at the GW170817-bound normalization ($D(\partial_{\hat{n}}\phi)^2 \lesssim 2 \times 10^{-15}$ path-averaged). The deformation is direction-even ($(\partial_{\hat{n}}\phi)^2$ is invariant under $\hat{n} \to -\hat{n}$), so it appears as a propagation residual between distinct paths rather than a same-path holonomy — the even-parity channel of the same $B(\phi)$ sector the corpus's holonomy program bounds on the odd/non-exact channel. On the corpus's weak-field branch $B(\phi) \simeq B_0\varphi^2$ (Paper 0, §2.2), the product $D(\partial_{\hat{n}}\phi)^2$ carries the quadratic field-depth suppression, so the stated ceiling is the saturation value of the GW170817 bound rather than the envelope's expectation on weak-field halo paths: the corpus-calibrated admissible normalization lies parametrically below it. The falsification threshold therefore bounds the normalization — not merely tests an assumed saturation — and the conversion into a constraint on the dimensionful B_0 belongs to the same open normalization map as the corpus's B-sector bookkeeping (Paper 0, Appendix E).
- **Constraint basis:** GW170817 bounds the path-averaged monopole $|c_\gamma - c_g|/c \lesssim 10^{-15}$; the conformal sector contributes exactly zero propagated inter-image delay by null-cone invariance (§3.1.3).
- **Delay scale:** $\Delta\tilde{\tau} \sim 10^{-3}$--1 s (milliseconds to seconds) on halo scales — $\Delta\tau_B \lesssim 0.2$ s per 2 Mpc halo path at the bound — the only propagating inter-image delay the theory supplies.
- **Primary observables:** Time-domain signatures in rapidly varying sources—lensed FRBs, GRBs.
- **Dark matter status:** TEP is a *precision systematic* in time-delay cosmography, not a wholesale DM replacement.
- **Falsification:** Null detection of achromatic timing residuals at < 0.1 ms on a 2 Mpc halo path bounds the path-averaged disformal product $D(\partial_{\hat{n}}\phi)^2 \lesssim 5 \times 10^{-19}$ — ~ 3.5 orders of magnitude below the GW170817 level — probing between the corpus-calibrated envelope and the saturated ceiling: the same $B(\phi)$ instrument as the holonomy program, applied to the even-parity multipath channel.

Regime II: the bookkeeping-inclusive case

- **Assumption (Sector Decoupling):** GW170817 constrains only differential disformal coupling; common-mode conformal temporal structure is not directly constrained by it, though it remains bounded by clock, PPN, and redshift channels (see Axiom 4).
- **Clock-transfer scale:** $\Delta\tilde{\tau} \sim$ 1--10 years, driven by the conformal factor $A(\phi)$ integrated over halo scales (Mpc). This is the *reconstruction-space* scale — the reconstruction-space clock-transfer map of §2.2 — not a propagated photon delay; the conformal contribution to any inter-image arrival-time residual is exactly zero, and the propagating differential channel remains the disformal carrier at the ms–s level.
- **Primary observables:** The reconstruction-space and time-domain content of the conformal sector — chronometric bookkeeping, endpoint clock-transfer maps, and the source-dependent temporal-composite response — together with the source-independent unstretched-space temporal refraction ($\tilde{g}_{\mu\nu} = -A^2_{\rm loc}(u_{\rm grav}(\mathbf{x}))U_\mu U_\nu + A^2_{\rm sp}(u_{\rm drift}(t,\mathbf{x}))\Pi_{\mu\nu}$, with $U_\mu \equiv -\nabla_\mu\bar{\phi}/\sqrt{2\bar{X}}$, rescaling the ADM spacetime lapse $N_{\rm TEP} = A(\phi)N_{\rm GR}$ at fixed spatial 3-metric γ_{ij}) and the *dynamical–lensing mass difference* in matched spatial observables. While isotropic conformal rescaling $\tilde{g}_{\mu\nu} = A^2(\phi)g_{\mu\nu}$ cancels out of null rays and leaves only the Planck-suppressed scalar stress tensor ($\kappa_\phi \simeq 6.5 \times 10^{-6}$; Paper 19, Step 64), the unstretched-space temporal metric retains intermediate clock rates $A(\phi(x))$ along every photon path, reproducing the Bullet Cluster galaxy-peak convergence (Section 6.9, Step 02), the SN Refsdal Fermat delay (Paper 19, Step 64), and the Planck CMB lensing cosmic-web amplification (Paper 18, Step 21) from baryons and the selected two-branch action $P_{,X} = k\sqrt{\xi} + 2\xi$.
- **Dark matter status:** Phantom Mass is the apparent mass component inferred when temporal structure is reconstructed under the Isochrony Axiom: for dynamics it reflects the Temporal Shear acceleration on slow matter; for static-source lensing across unstretched space it reflects the line-of-sight refractive index $n(x) \simeq 1 - 2\Phi_N(x)/c^2 - \beta_A u_{\rm grav}(x)$ of intermediate temporal depth; and for temporally structured sources it includes the bounded projected image response ($\lesssim 10^{-7}$).
- **Falsification:** The unstretched-space temporal refraction channel is tested directly against cluster convergence morphology (Bullet Cluster galaxy-peak alignment and absence of a negative moat, Step 02), strong-lensing Fermat delays (SN Refsdal, Paper 19 Step 64), and the CMB lensing spectrum (Paper 18 Step 21), while source-variability correlations test the bounded temporal-composite contribution.

This work demonstrates that *Regime II operates as a time-domain and reconstruction-space framework* alongside the source-independent optical refraction of the unstretched-space temporal metric (§6.9). Regime I is the conservative baseline in which the

spatial-gradient disformal deformation is evaluated at its GW170817-bounded normalization, while under the unstretched-space temporal metric $\tilde{g}_{\mu\nu} = -A^2_{\rm loc}(u_{\rm grav}(\mathbf{x}))U_\mu U_\nu + A^2_{\rm sp}(u_{\rm drift}(t,\mathbf{x}))\Pi_{\mu\nu}$ intermediate temporal depth $-\beta_A u_{\rm grav}(x)$ along the photon path closes the source-independent deflection across galaxy, cluster, and CMB lensing while preserving $c_{\rm GW} = c_\gamma$ identically on the shared temporal-lapse cone.

2.7 Why Lensing May Not Be Purely Spatial

Standard gravitational lensing analysis contains a simplifying temporal assumption that has typically been treated as exact. This assumption is made explicit, and its breakdown under TEP leads directly to the dark matter reinterpretation.

The Hidden Assumption in Image Formation

When a lensed galaxy is observed, photons are collected over an exposure time and an "image" is reconstructed. This reconstruction implicitly assumes:

1. All photons arriving during the exposure left the source at approximately the same epoch
2. Differences in arrival angles correspond to differences in spatial paths, not temporal paths
3. The reconstructed "shape" represents a spatial snapshot of the source at one moment

These assumptions constitute the *Isochrony Axiom* applied to image formation. Standard analysis corrects for the mean lookback time (distance), but assumes that the variance in lookback time across the image is negligible. This overlooked variance is defined as *Differential Lookback Time*. In a two-metric framework, this variance is an additional source-dependent reconstruction residual.

The temporal composite Mechanism

Consider an extended galaxy being lensed. Light from different parts of the galaxy:

- Leaves the source at different times (the source is evolving on Myr timescales)
- Takes different paths through the lens (different impact parameters)
- Is accompanied by a different matter-clock transfer congruence through regions with different $A(\phi)$ values — a timelike reconstruction field, not a photon propagation delay
- Arrives at the detector at the "same" observation time

If $A(\phi)$ varies spatially—forming a halo-like configuration around the lens—then the matter-frame path time

$$\Delta t_{\rm path} = \int_{\rm path} \frac{A(\phi)}{c}\, dl$$

evaluated on the associated timelike clock congruence differs between rays at different impact parameters. This functional is not a photon arrival time: Axiom 1 makes the photon coordinate transit conformally invariant, so no part of it propagates to any observed inter-image delay. What it quantifies is the reconstruction-space contrast — the magnitude of the clock-transfer field that a universal-clock reconstruction would misassign to mass, distance, or time-delay normalization. For halo-scale propagation depths $L \sim 2$ Mpc and conformal variations $\Delta A/A \sim 10^{-6}$, that envelope contrast is:

$$\Delta\tilde{\tau}_{\rm env} \sim \frac{\Delta A}{A}\cdot\frac{L}{c} \sim \text{years}$$

The propagating residual on the same paths is exactly zero in the conformal sector (Paper 19, step_56) and is bounded at the ms–s level on the disformal carrier (Box 3.2). The relevant comparison for any claimed observable is therefore the post-model residual — constrained to $\lesssim$ tens of days in existing strong-lens systems by the success of GR mass models (Paper 19) — not the multi-hundred-day total delay, which those models already reproduce. For slowly evolving galaxies the source-dependent temporal-composite response is negligible. The dominant signal is therefore the coherent static temporal-field sector: physical optical shear/convergence arises from scalar backreaction and any permitted disformal response, while the conformal sector supplies the associated chronometric reconstruction. The temporal smearing (the source-motion term) becomes the dominant propagating temporal-composite signal only for fast transients (FRBs); for slowly evolving sources its observable content is the shear-variance residual above the standard delay-map baseline (§3.4).

Why This Creates "Phantom Mass"

For a source that evolves on timescales comparable to the emission-time-map contrast $\Delta T_{\rm eff}$:

- If the source was more compact in the past → inner regions (earlier epoch) appear smaller

- If the source was less compact in the past → inner regions appear larger
- The reconstructed "shape" is systematically distorted by temporal mixing

Standard lensing analysis interprets *any* systematic shape distortion as evidence for mass (convergence and shear). The temporal smearing is mathematically indistinguishable from gravitational lensing distortion, but its amplitude is bounded by the ratio of the emission-time-map contrast to the source-evolution timescale: $\lesssim 10^{-7}$ in shear for galaxy sources, vanishing for static sources. It is therefore a source-dependent reconstruction signature, not the carrier of the observed dark-matter-level lensing excess; that excess is confronted through the dynamical–lensing mass comparison (§2.6).

Why This Effect Is Achromatic

The conformal factor $A(\phi)$ rescales proper time identically for all photon frequencies. Unlike plasma dispersion (which scales as ν^{-2}) or dust extinction (wavelength-dependent), conformal temporal coupling is *perfectly achromatic*. General relativistic lensing is likewise achromatic, so achromaticity alone does not discriminate a temporal-field contribution from ordinary gravitational lensing. The operational significance is the converse: because the TEP contribution is achromatic, it cannot be separated from GR lensing by colour information, but it *can* be separated from plasma and dust systematics — which is precisely what makes multi-frequency timing the discriminating measurement (Section 5.1).

The Central Thesis

Under TEP, the existence of "dark matter" is inferred rather than directly observed. It is a *conclusion contingent on the Isochrony Axiom*. If that axiom fails—if the universe is not temporally synchronous in the way standard analysis assumes—then the inferred "dark mass" can be modeled as unmodeled temporal structure. The dark sector is modeled not as a substance, but as the shadow of time. No claim is made regarding uniqueness of the two-metric realization, only the operational equivalence of temporal gradients to inferred mass under isochrony.

3. The Phantom Mass Mechanism

3.1 The Matter-Metric Map and Null-Trajectory Structure

The Phantom Mass mechanism is derived from the TEP matter-metric architecture (Jakarta, Paper 0). In isotropic shorthand, the gravitational metric $g_{\mu\nu}$ is related to the causal matter metric $\tilde{g}_{\mu\nu}$ through the conformal–disformal map:

$$\tilde{g}_{\mu\nu} = A^2(\phi)\, g_{\mu\nu} + B(\phi)\, \nabla_\mu\phi\, \nabla_\nu\phi$$

where $A(\phi)$ is the universal conformal coupling and $B(\phi)$ is the disformal coupling. This isotropic conformal–disformal expression provides the historical benchmark (not a Newtonian-gauge metric with a conformal factor inserted only in the spatial sector), whereas the operative TEP realization adopts the preferred-foliation decomposition $\tilde{g}_{\mu\nu} = -A^2_{\rm loc}(u_{\rm grav}(\mathbf{x}))U_\mu U_\nu + A^2_{\rm sp}(u_{\rm drift}(t,\mathbf{x}))\Pi_{\mu\nu} + B(\phi)\Pi^\alpha_\mu\Pi^\beta_\nu\nabla_\alpha\phi\nabla_\beta\phi$ along the cosmic temporal flow $U_\mu \equiv -\nabla_\mu\bar{\phi}/\sqrt{2\bar{X}}$, as specified in Paper 0, Section 2.2 (§1.5, §6.9). Examining the isotropic benchmark first isolates why isotropic ruler-stretching cancels from null trajectories before evaluating the unstretched-space temporal refraction in §6.9.

3.1.1 Conformal Invariance of Null Trajectories

In the purely conformal subclass ($B = 0$), the matter metric reduces to $\tilde{g}_{\mu\nu} = A^2(\phi)\, g_{\mu\nu}$. A standard result of conformal geometry in four dimensions is that null geodesics are preserved: if k^μ is tangent to a null geodesic of $g_{\mu\nu}$, the same path is a null geodesic of $\tilde{g}_{\mu\nu}$ with a rescaled affine parameter. Therefore:

$$B = 0 \quad\Longrightarrow\quad d\tilde{s}^2 = 0 \quad\Longleftrightarrow\quad ds^2 = 0.$$

This has three immediate consequences for gravitational lensing:

1. A^2 alone does not change the null trajectory. Pure conformal rescaling preserves null cones; it cannot generate a spatial refractive index or bend photons onto new paths.
2. Static geometric bending beyond the conformally related geometry must come from $g_{\mu\nu}[\phi]$ backreaction — the scalar field modifying the Einstein-frame metric through the coupled field equations — or from $B \neq 0$, which tilts the matter null cone relative to the gravitational null cone.
3. The conformal sector can alter endpoint clock calibration, frequency transport, and timelike source evolution. It acts directly on timelike observables (orbital periods, spectroscopic velocities, standard-candle distance calibration) but cannot generate the

printed spatial refractive index $n_{\rm eff} \simeq 1 - 2\Psi + (A(\phi) - 1)$ that would follow from inserting $A^2(\phi)$ into the spatial sector of a Newtonian-gauge metric alone.

3.1.2 Separating Null and Timelike Observables

The lensing deflection angle decomposes into two physically distinct channels:

$$\delta\theta_{\rm lens} = \delta\theta_{g[\phi]} + \delta\theta_B$$

where $\delta\theta_{g[\phi]}$ is the deflection from scalar backreaction on the geometric metric $g_{\mu\nu}$ — the scalar field modifying the Einstein-frame curvature through the coupled Einstein–scalar equations — and $\delta\theta_B$ is the deflection from the disformal term $B(\phi)\nabla_\mu\phi\nabla_\nu\phi$, which tilts the matter null cone along the field gradient. On the isotropic-conformal benchmark ($\tilde{g}_{\mu\nu} = A^2(\phi)g_{\mu\nu}$), the conformal factor $A(\phi)$ does not appear in $\delta\theta_{\rm lens}$ at leading order because it cancels from the null condition, whereas under the canonical unstretched-space temporal metric ($\tilde{\gamma}_{ij} = \gamma_{ij}$ at fixed cosmic epoch $A_{\rm sp} = 1$, §6.9), the local temporal lapse $A_{\rm loc}(u_{\rm grav})$ supplies the direct refractive deflection $\delta\theta_{A_{\rm loc}}$ via $n(\mathbf{x}) \simeq 1 - 2\Phi_N/c^2 - \beta_A u_{\rm grav}(\mathbf{x})$.

The worked backreaction example on the isotropic-conformal benchmark is the scalar-Gauss-Bonnet (sGB) coupling studied in Bahrain (Paper 28, Appendix L). In shift-symmetric sGB, the scalar field $\phi \sim Q_s/r$ backreacts on the geometric metric at $\mathcal{O}(\eta^2)$ (where $\eta = 3\alpha_{\rm GB}/M^2$), producing a genuine correction to the null-trajectory bending. Under isotropic conformal rescaling $A = e^{-\phi}$ modifies clock rates at $\mathcal{O}(\eta)$ — acting on timelike observables such as the ISCO — without contributing to the shadow at leading order because isotropic conformal transformations preserve null cones. Bahrain Appendix L explicitly isolates this sGB backreaction contribution and demonstrates the different coupling orders of the null and timelike observables on that benchmark: the shadow is sensitive to the geometric metric at $\mathcal{O}(\eta^2)$ while the ISCO feels the conformal factor at $\mathcal{O}(\eta)$.

3.1.3 The temporal-composite observable

The conformal sector produces observable effects through timelike channels: endpoint clock calibration, frequency transport, and source evolution. These are captured by the temporal composite observable, which relates the observed image intensity to the emission-time shift induced by the temporal field:

$$\delta I_{\rm obs} \simeq -\frac{\partial I_{\rm src}}{\partial t_{\rm em}}\,\delta t_{\rm em}$$

where $\delta t_{\rm em}$ is the differential emission-time shift between paths, sourced by the complete metric $\tilde{g}_{\mu\nu}$ rather than an unmatched spatial conformal factor. The delay must come from the full disformal metric — including $g_{\mu\nu}[\phi]$ backreaction, the disformal term B, endpoint clock-rate differences through $A(\phi)$, and frequency transfer — not from a path integral $\int A(\phi)\,dl/c$ interpreted as photon proper-time accumulation. Null proper time is zero (Jakarta); any measured open-path effect must be formulated through emission and reception clocks, actual coordinate delay from the geometric and disformal metric, or frequency transfer.

The static clock-transfer contribution from the conformal sector modifies the relation between accumulated matter-frame time and the geometry reconstructed under an isochronous model. The associated effective clock-transfer contrast along a path γ is:

$$\Delta\tilde{\tau}_{\rm static} = \frac{1}{c}\int_\gamma \big(A(\phi) - 1\big)\,dl$$

This is a clock-transfer discrepancy — a difference between the matter-frame time a timelike congruence accumulates along each path and the time an isochronous GR lens model infers — not a propagated photon delay under isotropic conformal rescaling ($\tilde{g}_{\mu\nu} = A^2(\phi)g_{\mu\nu}$). On the isotropic-conformal benchmark its differential photon content vanishes identically: the null condition is conformally invariant (§3.1.1), and the underlying Temporal Shear is an exact one-form, $\Sigma_\mu = \nabla_\mu \ln A$, whose open-path integral is endpoint-determined and cancels between any two rays joining the same source and observer; inserting the screening projection $\mathcal{S}_\Sigma(\mathcal{E})$ inside a photon-path observable is a category error, since photons carry no clocks and do not respond to the shear sector. The dedicated strong-lensing pipeline evaluates the exact additional static isotropic-conformal residual at 0.0 d (Paper 19, step_56). On that isotropic benchmark the functional is retained as the chronometric reconstruction map — the field a universal-clock reconstruction would misassign to convergence, mass-sheet normalization, or delay calibration — and it contributes no term to the propagated Fermat surface, while the spatial-disformal multipath deformation is bounded by GW170817 to $\lesssim 0.2$ s per 2 Mpc halo path. By contrast, under the canonical unified unstretched-space temporal metric $\tilde{g}_{\mu\nu} = -A^2_{\rm loc}(u_{\rm grav}(\mathbf{x}))U_\mu U_\nu + A^2_{\rm sp}(u_{\rm drift}(t,\mathbf{x}))\Pi_{\mu\nu} + B(\phi)\Pi^\alpha_\mu\Pi^\beta_\nu\nabla_\alpha\phi\nabla_\beta\phi$ ($U_\mu \equiv -\nabla_\mu\bar{\phi}/\sqrt{2\bar{X}}$; Paper 0 §2.2, Step 80, evaluated in §6.9 below), where the $(3+1)$ ADM lapse $N_{\rm TEP} = A_{\rm loc}(u_{\rm grav})N_{\rm GR}$ rescales along the cosmic temporal flow while

the spatial 3-metric $\tilde{\gamma}_{ij} = \gamma_{ij}$ remains unstretched at fixed epoch, intermediate clock rates $A_{\rm loc}(u_{\rm grav}(\mathbf{x}))$ do not cancel against spatial stretching and refract null rays directly via $n(x) \simeq 1 - 2\Phi_N(x)/c^2 - \beta_A u_{\rm grav}(x)$.

3.1.4 The Amplification Matrix and Jacobian Decomposition

Integrating the geodesic deviation equation along the line of sight yields the amplification matrix $\mathcal{A}_{ij}$, which maps source-plane displacements to image-plane displacements:

$$\mathcal{A}_{ij} = \frac{\partial \beta_i}{\partial \theta_j} = \delta_{ij} - \int_0^{\chi_s} \frac{(\chi_s - \chi)\chi}{\chi_s}\, \mathcal{R}_{ij}(\chi)\, d\chi$$

where $\mathcal{R}_{ij}$ is the optical tidal matrix constructed from the Riemann tensor of the geometric metric $g_{\mu\nu}[\phi]$, projected onto the screen space. On the isotropic-conformal benchmark ($\tilde{g}_{\mu\nu} = A^2(\phi) g_{\mu\nu}$), the TEP correction to the optical tidal matrix comes from scalar backreaction on $g_{\mu\nu}$ and from the disformal sector, not from a spatial conformal factor:

$$\mathcal{R}_{ij} = \mathcal{R}_{ij}^{(\Psi)} + \mathcal{R}_{ij}^{(g[\phi])} + \mathcal{R}_{ij}^{(B)}$$

where $\mathcal{R}_{ij}^{(\Psi)}$ is the standard GR contribution, $\mathcal{R}_{ij}^{(g[\phi])}$ is the scalar-backreaction correction to the geometric metric, and $\mathcal{R}_{ij}^{(B)}$ is the disformal correction. Under isotropic conformal rescaling, the conformal factor does not appear at leading order because it cancels from the null trajectory; under the canonical unstretched-space temporal metric ($\tilde{\gamma}_{ij} = \gamma_{ij}$ at fixed cosmic epoch, §6.9), by contrast, the local temporal lapse $A_{\rm loc}(u_{\rm grav})$ adds the direct refractive tidal term $\mathcal{R}_{ij}^{(A_{\rm loc})} = \beta_A \partial_i \partial_j u_{\rm grav}$.

The magnification of an image is $\mu = (\det \mathcal{A})^{-1}$. To first order in the TEP correction $\delta\mathcal{A}_{ij}$, the fractional change in magnification is:

$$\frac{\delta\mu}{\mu} = \mathrm{Tr}\left(\mathcal{A}_{\rm GR}^{-1}\, \delta\mathcal{A}\right)$$

This defines the lensing amplification kernel $\mathcal{P}_\mu$, a projection operator that maps the scalar-backreaction and disformal Hessian to the observable magnification shift. Near a critical curve, where $\det \mathcal{A}_{\rm GR} \to 0$, the kernel diverges as $\mu_{\rm GR}^2$, amplifying small corrections into large observable residuals. Because the inverse reconstruction is correspondingly sensitive to small timing and geometric perturbations there, this supplies a candidate mechanism capable of closing the factor-of-$\sim$90–750 amplitude gap between direct potential-sampling and observed delay shifts identified in Paper 19 (SN Refsdal) — locating that discrepancy in Jacobian amplification rather than in a free phenomenological coefficient. A numerical resolution requires evaluation of the complete tensor and lens-model response, not the scalar log-magnification proxy alone; that calculation is not performed here.

Formally, the operational response proxy used in the blind-prediction tests (Paper 19, §2.3) is the first-order truncation of this kernel:

$$\Gamma_t(i) = 1 + \kappa_{\rm lens}\, \mathcal{P}_\mu\big[\nabla\phi\big](i) \;\approx\; 1 + \kappa_{\rm lens} \log_{10}\big(\mu_{\rm norm}(i)\big)$$

This is a phenomenological scalar truncation of the tensor response. Weak shear alone does not make it exact; exact reduction additionally requires an approximately isotropic response, $\delta\mathcal{A}_{ij} \propto \delta_{ij}$. The log-magnification form follows from $\delta\mu/\mu = \mathrm{Tr}(\mathcal{A}^{-1}\delta\mathcal{A})$ when $\delta\mathcal{A}$ is proportional to the identity. The mu–kappa–gamma systematic is the residual error from truncating the full tensor kernel to a scalar log-magnification proxy. Eliminating it requires direct evaluation of $\mathcal{P}_\mu$ from high-resolution mass models, which is the definitive next phase identified in Paper 19.

Response-coefficient convention. Following the corpus rule established in TEP-COS (Paper 10) for $\kappa_{\rm MSP}$, the quantity $\kappa_{\rm lens}$ is an *observable channel response coefficient*: it measures how strongly the lensing reconstruction responds to Temporal Shear. It is not the bare microscopic conformal coupling β_A, not $A - 1$, and not the locally active PPN coupling, and it must not be numerically identified with the amplitudes of other channels — such as $\epsilon_T^{\rm HC}$ (TEP-HC) or the line-of-sight amplitude of TEP-C0 — unless a solved environmental transfer function establishes that identification. Schematically, the lensing-channel observable takes the form $\Delta O_{\rm GL} = \kappa_{\rm GL}\, \mathcal{S}_{\rm GL}(\mathcal{E})\, \mathcal{F}_{\rm GL}[\Delta \ln A, \Sigma_\mu, C_A;\ \Phi, \rho, z]$, with the environmental screening operator applied before comparison with any other channel.

Amplitude dictionary. The manuscript refers to several distinct quantities that must not be conflated:

- β_A: the microscopic conformal parameter (bare Lagrangian quantity).
- $\Sigma_\mu = \nabla_\mu \ln A(\phi)$: the underlying Temporal Shear.
- $\Sigma_\mu^{\rm obs} = \mathcal{S}_\Sigma(\mathcal{E})\,\Sigma_\mu$: the observable environmentally projected Temporal Shear.
- $\langle \Delta A \rangle_\gamma = \int_\gamma \Sigma_\mu^{\rm obs}\, dx^\mu$: the open-path clock-transfer contrast. In the unscreened limit this is the integral of an exact one-form, $\int_\gamma \nabla_\mu \ln A\, dx^\mu = \ln A_{\rm obs} - \ln A_{\rm em}$, so it is endpoint-determined and its differential between two rays sharing the same endpoints vanishes identically; the environment-projected form is defined on the matter-clock congruence and inherits the same vanishing multipath photon content.
- $\kappa_{\rm lens}$: the lensing reconstruction response coefficient relating temporal-field structure to inferred convergence.

These quantities are not numerically interchangeable without a solved environmental transfer function: on the isotropic-conformal benchmark the conformal entries act on clock calibration and reconstruction while the propagating delay field is the disformal multipath deformation bounded in Box 3.2, whereas under the canonical unstretched-space temporal metric (§6.9, Step 02; Paper 19, Step 64) the local temporal depth $-\beta_A u_{\rm grav}(\mathbf{x})$ along the shared ADM lapse directly supplies the static optical Fermat potential and deflection.

3.1.5 Interpretation: Geometric vs. Temporal Contributions

The Jacobian decomposition reveals two physically distinct contributions to image distortion:

Term	Source	Physical Origin	Observational Signature
Geometric	$\Psi_{,ij}$	Spatial curvature from mass	Standard convergence κ and shear γ
Scalar backreaction	$g_{\mu\nu}[\phi]$	Scalar field modifying Einstein-frame curvature	Coherent tangential shear mimicking DM halo
Disformal	$B(\phi)\nabla_\mu\phi\nabla_\nu\phi$	Null-cone tilt along field gradient	Direction-dependent deflection
temporal composite	$\mu_s^i\,\partial_j \Delta T_{\rm eff}$	Source motion × emission-time-map gradient	Kinematics-correlated shear variance; the residual above the standard delay-map baseline is the TEP-specific content

The scalar-backreaction and disformal terms produce coherent contributions to the shear field. The temporal composite term arises when the source position evolves during the differential emission-time contrast $\Delta T_{\rm eff}$ across the image; for a source with proper motion $\vec{\mu}_s$, the effective source position becomes:

$$\beta^i_{\rm eff}(\vec{\theta}) = \beta^i_{\rm geom}(\vec{\theta}) - \mu_s^{\,i}\,\Delta T_{\rm eff}(\vec{\theta})$$

where $\mu_s^{\,i}$ is the i-th component of the source proper motion and $\Delta T_{\rm eff}$ is the effective emission-time map of §2.5. The corresponding temporal-composite contribution to the Jacobian is:

$$\delta\mathcal{A}^{\rm TC}_{ij} = -\mu_s^{\,i}\,\partial_j \Delta T_{\rm eff}$$

This adds an asymmetric, source-dependent contribution to the Jacobian that does not average coherently but increases the variance of shear measurements.

Summary: The Phantom Mass Decomposition

The full amplification matrix in TEP is:

$$\mathcal{A}_{ij} = \underbrace{\mathcal{A}_{ij}^{\mathrm{GR}}}_{\text{Baryonic Lensing}} + \underbrace{\mathcal{A}_{ij}^{(g[\phi],\mathrm{static})}}_{\text{"Dark Matter" (Coherent)}} + \underbrace{\mathcal{A}_{ij}^{(\mathrm{dyn})}}_{\text{temporal composite (Stochastic)}}$$

Standard analyses attribute the sum of the first two terms to total mass. TEP identifies the second term as the coherent optical component of the deflection ledger and evaluates both metric realizations from the action: under isotropic conformal rescaling ($\tilde{g}_{\mu\nu} = A^2 g_{\mu\nu}$, where spatial stretching cancels $A^2(\phi)$ out of null rays), Einstein-frame scalar stress-energy reaches $\kappa_\phi \approx 4 \times 10^{-5}$ on the unit-term control ($\simeq 6.5 \times 10^{-6}$ on the selected two-branch sector; Paper 19, Steps 61, 64) and the spatial-disformal term $B(\partial\phi)^2$ is bounded $\sim 5\times$–$70\times$ below the cluster-lensing requirement (Step 63; §6.9), whereas under the canonical unified unstretched-space temporal metric $\tilde{g}_{\mu\nu} = -A_{\mathrm{loc}}^2(u_{\mathrm{grav}}(\mathbf{x}))U_\mu U_\nu + A_{\mathrm{sp}}^2(u_{\mathrm{drift}}(t,\mathbf{x}))\Pi_{\mu\nu}$ (§6.9, Step 02; Paper 19, Step 64; Paper 18, Step 21), intermediate temporal depth $-\beta_A u_{\mathrm{grav}}(x)$ along the shared ADM lapse refracts photons via $n(x) \simeq 1 - 2\Phi_N/c^2 - \beta_A u_{\mathrm{grav}}(x)$ and reproduces the Bullet Cluster convergence, SN Refsdal Fermat delay, and Planck CMB lensing power from baryons alone. The third term supplies the source-dependent discriminating channel: shear variance correlated with source kinematics. The correlation's existence is not itself distinctive, since any emission-time map sources the same $\mu_s\,\partial_j\Delta T$ structure — under particle dark matter and GR a moving source's effective source-plane position is $\beta - \vec{\mu}_s\Delta T$, and the ordinary Fermat surface alone produces $\gamma^{\mathrm{std}} = \mu_s\,\partial_\theta \Delta T_{\mathrm{Fermat}} \sim 3 \times 10^{-8}$ for strong-lens geometries ($\sim 6 \times 10^{-10}$ at galaxy–galaxy scales; step_01). The discriminating observable is the residual over that computable baseline: on the spatial-disformal carrier it is bounded at $\gamma^{\mathrm{TC}}_{\mathrm{prop}} \sim 10^{-15}$ — seven orders below the standard baseline and tested directly by the timing residuals of §5.1 — while the reconstruction-space contribution is bound-set rather than measured by the scatter test of §3.4. On the isotropic-conformal benchmark $A^2(\phi)g_{\mu\nu}$, $A(\phi)$ acts on timelike observables and clock calibration but generates neither null-trajectory bending nor any inter-image arrival-time residual (exact zero in Paper 19, step_56), whereas on the unstretched-space temporal metric (§6.9) the lapse $A_{\mathrm{loc}}(u_{\mathrm{grav}})$ directly refracts null trajectories across unstretched space.

Box 3.1: A Minimal Toy Model Estimate (Halo Scale Integration)

To demonstrate the order of magnitude, consider a simple spherical conformal halo profile, applied as a local interior model over the field-bearing extent of the halo:

$$A(\phi) = 1 + \epsilon \ln(r/r_0), \qquad r \lesssim r_h \sim L_{\rm halo}/2$$

For a coupling strength $\epsilon \approx 10^{-6}$ and a characteristic scale $r_0 = 10$ kpc, the excursion $|A(\phi) - 1|$ remains at the 10^{-6} level throughout the domain ($|\ln(r/r_0)| \lesssim 5$ for $r \in [r_E, r_h]$). The ambient boundary condition is imposed by environmental matching rather than by the toy form: beyond the halo matching radius r_h the screened field relaxes to the ambient baseline $A_\infty = 1$, so the asymptotic divergence of the logarithm lies outside the domain of the estimate and is never extrapolated.

- **Integration Path:** The delay is integrated only over the effective halo depth ($L_{halo} \approx 2$ Mpc), not the full cosmological path. This respects the locality of the potential well.
- **Chronometric-Envelope Contrast:** Across an Einstein radius ($r_E \approx 5$ kpc), the differential clock-transfer field corresponds to:

$$\Delta\tilde{\tau}_{\rm env} \sim \frac{\epsilon}{2}\frac{L_{halo}}{c} \approx \frac{10^{-6}}{2} \cdot (6.5 \times 10^6 \text{ light-years}) \approx 3.2 \text{ years}$$

- **Carrier status:** This ~3 year figure is the reconstruction-space scale — the reconstruction-space clock-transfer field of §2.2 and §3.1.3 — not a propagated photon delay under isotropic conformal rescaling ($\tilde{g}_{\mu\nu} = A^2 g_{\mu\nu}$). On the isotropic-conformal benchmark the exact additional static-conformal inter-image residual on the same paths is 0.0 d (Paper 19, step_56), the propagating differential residual is carried by the disformal multipath deformation at $\lesssim 0.2$ s per 2 Mpc, and scalar backreaction enters the delay surface only as ordinary g-sector Fermat structure; by contrast, under the canonical unstretched-space temporal metric ($\tilde{\gamma}_{ij} = \gamma_{ij}$, §6.9; Paper 19, Step 64), intermediate temporal depth $-\beta_A u_{\rm grav}(\mathbf{x})$ on the selected two-branch sector $P_{,X} = k\sqrt{\xi} + 2\xi$ supplies the multi-hundred-day static Fermat delay directly ($+330.9$ d sheet and $+221.6$--443.2 d 3-D NFW ray on SN Refsdal from $f_b = 0.16$ baryons alone). The ~3 year chronometric-envelope figure bounds the post-model residual an isochronous reconstruction could misassign, corresponding to the residual class measured by Paper 19 ($\lesssim$ tens of days).
- **Clock-transfer reconstruction:** The static gradient $\nabla(\Delta\tilde{\tau}_{\rm env})$ defines the unmodeled clock-transfer field along existing lens paths. When a GR lens model is required to reproduce a path-dependent timing map while assuming isochrony, such a residual would be absorbed into inferred convergence, mass-sheet normalization, or distance calibration — the reconstruction-space "Phantom Mass" signature. Under isotropic conformal rescaling ($\tilde{g}_{\mu\nu} = A^2 g_{\mu\nu}$) it does not refract null geodesics onto new trajectories and does not shift propagated arrival times because isotropic conformal rescaling preserves null cones; under the canonical unstretched-space temporal metric (§6.9, Step 02), however, intermediate temporal depth $A_{\rm loc}(u_{\rm grav})$ does not cancel against spatial stretching and supplies direct optical convergence and Fermat delay alongside the scalar-backreaction and disformal channels of §3.1.4.
- **Source-motion term (stochastic):** The term $\vec{\mu}_s \cdot \nabla(\Delta T_{\rm eff})$ couples source proper motion to whichever emission-time map propagates. On the propagating carrier it is the millisecond-to-second disformal contribution, which dominates the differential delay budget for fast transients (the lensed-FRB channel of §5.1); for slowly evolving galaxy sources its observable content is the shear-variance residual above the standard-map baseline, bounded per Box 3.2.

Profile Dependence Check: The ~3 year estimate uses a logarithmic profile for simplicity. Realistic dark matter halos follow the NFW profile:

$$\rho_{NFW}(r) = \frac{\rho_s}{(r/r_s)(1 + r/r_s)^2}$$

If the clock-transfer field tracks the gravitational potential ($A(\phi) - 1 \propto \Psi$), then $A(\phi) - 1 \propto \int \rho/r \, dr$, giving:

$$(A(\phi) - 1)_{\rm NFW}(r) \propto \ln(1 + r/r_s) - \frac{r/r_s}{1 + r/r_s}$$

For a cluster with $r_s \approx 200$ kpc and integration over $L_{halo} \approx 2$ Mpc:

- The NFW profile concentrates more of the clock-transfer field near the core than the logarithmic profile.
- The differential envelope contrast across an Einstein radius ($r_E \approx 5$--50 kpc) is enhanced by a factor of 2–5 relative to the logarithmic estimate.
- Result: $\Delta\tilde{\tau}_{\mathrm{env},NFW} \sim 5$--$15$ years at the reconstruction-space scale — a reconstruction-scale figure with the carrier status stated above, not an observed-delay prediction.

The order-of-magnitude envelope estimate is robust to profile shape; realistic NFW profiles produce slightly larger chronometric contrasts than the toy logarithmic model.

Box 3.2: Order of Magnitude Estimate for stochastic shear

To estimate the magnitude of the *stochastic* shear contribution (the dynamic term $\mu_s \nabla(\Delta\tilde{\tau})$), the analysis uses the updated halo-scale delays:

- **Source Velocity:** Typical cluster transverse velocity $v_s \sim 1000$ km/s at distance $D_A \sim 1$ Gpc yields an angular proper motion $\mu_s \approx 2 \times 10^{-7}$ arcsec/year ($\sim 0.2\,\mu$as/yr; the milliarcsec-scale figure appearing in earlier drafts reflected a unit slip).
- **Envelope Delay Gradient:** From Box 3.1, a reconstruction-space contrast of ~ 3 years varying over arcsecond scales gives $\nabla(\Delta\tilde{\tau}_{\mathrm{env}}) \sim 3$ years/arcsec at the envelope; the propagating-carrier gradient is 12 orders smaller (below).
- **Resulting stochastic shear:** The product is dimensionless shear:

$$\gamma_{stoch}^{\mathrm{env}} \approx (2 \times 10^{-7}\,\mathrm{arcsec/yr}) \times (3\,\mathrm{yr/arcsec}) \approx 6 \times 10^{-7}$$

This stochastic contribution ($\gamma_{stoch} \sim 10^{-6}$ at the reconstruction-space scale) is small compared to typical weak lensing shear ($\gamma \sim 0.01$--0.1), confirming that the dynamic term is a perturbation (excess scatter), not the dominant signal. The coherent "Dark Matter" halo signature is the combined static temporal-field contribution as registered by the lens model: under the isotropic-conformal benchmark its physical optical-tidal component is carried by scalar backreaction on $g_{\mu\nu}$ and any bounded disformal term while the conformal sector modifies the associated clock-transfer and inference mapping (the clock-transfer reconstruction), whereas under the canonical unstretched-space temporal metric (§6.9, Step 02) the local temporal lapse $A_{\mathrm{loc}}(u_{\mathrm{grav}})$ directly refracts null rays across the unstretched spatial manifold via $n(\mathbf{x}) \simeq 1 - 2\Phi_N/c^2 - \beta_A u_{\mathrm{grav}}(\mathbf{x})$. In neither case does the static halo convergence arise from source motion.

Two scales, two roles. The two delay scales of Box 3.3 sit on different carriers, and the applications of Section 5 must be read against the correct one. On the isotropic-conformal benchmark ($\tilde{g}_{\mu\nu} = A^2(\phi) g_{\mu\nu}$), the conformal clock-transfer map contributes zero differential inter-image delay: isotropic conformal rescaling preserves null cones (§3.1.1), so $A(\phi)$ drops out of the photon transport equation and the open-path integral is endpoint-determined (Paper 19 verifies the vanishing static-conformal residual explicitly), whereas under the canonical unstretched-space temporal metric ($\tilde{\gamma}_{ij} = \gamma_{ij}$ at fixed cosmic epoch, §6.9), intermediate temporal depth $-\beta_A u_{\mathrm{grav}}(x)$ supplies the static Fermat delay directly (Paper 19, Step 64). For the spatial-disformal multipath deformation $B(\phi)\nabla_\mu\phi\nabla_\nu\phi$, the cone-tilt carrier is bounded by GW170817 to $\lesssim 10^{-15}$ fractional — $\Delta T \lesssim 0.2$ s over the 2 Mpc halo path, the conservative-baseline scale. On that spatial-disformal carrier the same product gives $\gamma_{\mathrm{prop}}^{\mathrm{TC}} \sim \mu_s\, \partial_\theta \Delta T \sim (2 \times 10^{-7}\,\mathrm{arcsec/yr})(6.5 \times 10^{-9}\,\mathrm{yr/arcsec}) \sim 10^{-15}$. The 6×10^{-7} figure is the *reconstruction-space* envelope value: it quantifies the field a lens model would absorb when the clock-transfer map or unstretched-space Fermat surface is differentiated across a moving source, and it bounds the shear-variance applications of §5. The photon-propagating channel is tested directly and most sharply by the arrival-time residual itself (§5.1).

3.2 Connection to Lens-Model Degeneracies

This mechanism parallels the *Source-Position Transformation (SPT)* (Schneider & Sluse 2013), which identifies a degeneracy class of mass models yielding identical observables but differing time delays. TEP extends this degeneracy into the metric sector itself. Rather than permuting the mass distribution $\kappa(\vec{\theta})$ while holding the metric constant, TEP holds the baryonic mass constant and permutes the spacetime arrival surface $\Delta\tilde{\tau}(\vec{\theta})$. Consequently, the "Phantom Mass" can be understood as a physical realization of the SPT, where the extra freedom resides in the time-transport sector rather than invisible spatial matter.

3.3 The Two Regimes: Parameter Thresholds

TEP phenomenology divides into two distinct regimes, distinguished by the magnitude of the differential clock-transfer residual and the resulting phenomenology:

Box 3.3: Regime Definitions and Parameter Thresholds

Regime	Clock-transfer scale	Phenomenology
conservative baseline	ms–s (propagating, disformal carrier)	Time-domain residuals; static optical lensing effectively unchanged
Regime II	1--10 years (reconstruction space; propagating channel unchanged at ms–s)	Coherent scalar-metric/disformal lensing combined with chronometric reconstruction and source-dependent temporal-composite response

For a 2 Mpc path, the Regime-II envelope contrast is $\langle \Delta A \rangle_\gamma \sim 10^{-7}$--$10^{-6}$, consistent with the halo-scale integration in Box 3.1; per §3.1.3 this contrast is endpoint-determined for photons and is realized in the data only through the reconstruction mapping, while the propagating residual remains the disformal channel.

Operational Distinction: The conservative baseline accepts the standard GW170817 translation (arrival-time offset → propagation-speed bound) at face value. Regime II applies if TEP's dynamical-time interpretation is correct, in which case the standard translation may require revision—the observed $\Delta t = 1.74$ s constrains the *disformal* sector; it does not directly test the *conformal* sector, although conformal scalar sectors remain indirectly constrained by PPN, equivalence-principle, source-screening, and clock-comparison tests (Section 4).

Empirical Discriminator: The regime is determined by observation, not assumption. If lensed FRBs show only millisecond residuals, the conservative baseline applies. If strong-lens post-model residuals, reconstruction maps, or variability-correlated structure reveal clock-transfer structure at the envelope scale, Regime II is indicated.

Independent Regime Criterion (Breaking Circularity): To avoid circular reasoning, the following observable specifies the regime *independently* of TEP's correctness:

- **The Variability-Mass Correlation Test:** In Regime II, the inferred "dark matter" mass of a lens should correlate with the variability timescale of the background source population. Specifically: lenses observed through rapidly variable sources (AGN, quasars) should show systematically different mass reconstructions than the same lenses observed through slowly evolving sources (elliptical galaxies). This correlation is *forbidden* in standard CDM (mass is source-independent) but *required by the source-dependent temporal-composite component of Regime II*.
- **Decision Rule:** If existing strong-lens catalogs show no statistically significant correlation between inferred lens mass and source variability class at the >3σ level, the source-dependent temporal-composite component is disfavored. If such a correlation exists, it constitutes positive evidence for TEP independent of FRB timing. *Sensitivity floor:* the response enters through $\delta\kappa \sim \gamma^{\rm TC} = \mu_s\, \partial_\theta \Delta T_{\rm eff}$ (Box 3.2), so a correlation search at the proposed 10% mass precision probes $\gamma^{\rm TC} \gtrsim 10^{-1}$ — some five orders of magnitude above even the reconstruction-space value — and reaching the propagating carrier through mass reconstruction would require $\delta M/M \sim 10^{-15}$, far below any achievable precision. A null result at the specified precision therefore bounds the envelope amplitude rather than testing the carrier; the carrier itself is tested by the timing residual of §5.1. Complementarily, the ordinary delay-absorption route already constrains the propagating field: a path-dependent delay ΔT absorbed into a lens model shifts the inferred normalization by $\sim \Delta T/\Delta t_{\rm Fermat}$, so the observed $\sim 10\%$ closure of strong-lens mass models bounds any propagating residual at $\lesssim 10$ days per 100-day delay — excluding a propagating year-scale reading outright, consistently with the conformal null of §3.1.1.
- **Current Status:** This test can be performed with existing data (HST strong-lens archives, SDSS quasar lenses vs. galaxy-galaxy lenses) as an envelope-bounding correlation search; its discriminating power on the propagating carrier requires the timing channel of §5.1. This is flagged as a priority observational test.

Under the conservative baseline (Regime I), $\Delta\tilde{\tau}$ is small (milliseconds to seconds). In this regime:

- **Static Lensing:** The phantom mass effect is negligible for slowly evolving sources (galaxies). Static mass maps are unaffected.
- **Time-Domain Lensing:** The effect is dominant for fast transients. A millisecond gradient across an image plane is huge for an FRB.

Conclusion: In the conservative baseline (Regime I), TEP is a precision correction to time-domain astrophysics. In Regime II, TEP offers a conditional geometric reinterpretation in which the component conventionally attributed to dark matter may contain an unmodeled temporal-transport contribution.

3.4 The Critical Discriminator: Variability Scatter

Since the dynamic term $\mu_{s,i}\nabla_j(\Delta\tilde{\tau})$ is randomly oriented (depending on the direction of $\vec{\mu}_s$), it does not add to the mean shear profile but contributes to the *variance* of the shear measurement.

The Variability Scatter: The inferred "shear noise" (RMS dispersion of ellipticities) should be higher for source populations with high proper motion or intrinsic variability. While the coherent "dark matter" signal is static, the *scatter* around that signal is dynamic. The amplitude is set by the bookkeeping of Box 3.2: $\gamma^{\rm TC} \lesssim 6 \times 10^{-7}$ at the reconstruction-space scale and $\sim 10^{-15}$ on the propagating carrier, so the scatter excess contributes a variance $\gamma^2_{\rm TC}$ that is orders of magnitude below shape noise on either branch. The observable content of the test is therefore a bound on the reconstruction-space scale — a non-detection at a stated scatter precision caps $\nabla(\Delta\tilde{\tau})$ at that level — while the propagating channel is tested directly by timing residuals (§5.1).

The discriminating quantity is not the correlation's existence — the ordinary delay map produces kinematics-correlated dispersion at the standard level $\gamma^{\rm std} = \mu_s\, \partial_j \Delta T_{\rm Fermat}$ under particle dark matter as well (step_01: $\sim 3 \times 10^{-8}$ strong-lens, $\sim 6 \times 10^{-10}$ galaxy–galaxy) — but the residual above that baseline. Particle dark matter predicts equality with the standard-map value at every source class; a statistically significant excess bounds, and at sufficient precision reveals, a non-standard emission-time field. At achievable shear precisions the reach is the envelope-scale field and its reconstruction-space bound; the propagating disformal carrier sits ~7 orders below the baseline and is accessed by the timing channel of §5.1 rather than by shear.

4. Reanalysis of GW170817: What is Actually Constrained?

4.1 The Measurement and the Standard Translation

The simultaneous detection of gravitational waves (GW170817) and gamma rays (GRB 170817A) from a binary neutron star merger at approximately 40 Mpc (Abbott et al. 2017) represents one of the most precise measurements in astrophysics. The signals arrived within $\Delta t_{obs} = 1.74 \pm 0.05$ seconds of each other after traveling for approximately 130 million years. This is widely cited as constraining the difference between the speed of gravity c_g and the speed of light c_γ to (e.g., Baker et al. 2017; Creminelli & Vernizzi 2017; Ezquiaga & Zumalacárregui 2017; Sakstein & Jain 2017):

> Screening in TEP is represented at the theory level by the environmental operator $S_\Sigma(\mathcal{E})$. Quantities such as ρ_T, $R_T(M)$, $S_\oplus(r)$, compactness Φ/c^2, local stellar density, geometric coherence length, and channel-specific response coefficients are domain-specific projections of $\mathcal{E}$, not independent screening mechanisms and not interchangeable universal thresholds. Each is an observational transfer model that parameterizes the same underlying operator in a regime-appropriate form.

$$\frac{|c_\gamma - c_g|}{c} \lesssim 10^{-15}$$

This standard translation assumes that any delay is due to a uniform difference in propagation speed accumulating linearly over the entire cosmological distance. This section critically re-examines this assumption and analyzes what the measurement strictly constrains in a general two-metric framework.

4.2 The Common-Path Invariant: A Differential Measurement

A crucial but often overlooked geometric fact is that both messengers originated from the *exact same coordinate* in distant space. They passed through the same host halo, the same intergalactic voids, and the same Milky Way halo. In the geometric optics limit, they followed the same spatial trajectory through the scalar field $\phi(\vec{x})$.

The Common-Mode Cancellation:
Because the signals originate from the same coordinate and follow a nearly identical path, they do not diverge significantly. If the metric coupling contains a conformal component $\tilde{g}_{\mu\nu} = A^2(\phi) g_{\mu\nu}$, this factor rescales the common-mode endpoint clock and frequency calibration for *both* species identically. Any time dilation caused by passing through "time-warped regions" of space is experienced by both messengers. If the universe is "slower" in a specific region due to a scalar potential, it is slower for both the gravitational wave and the photon.

Implication for Conformal Coupling ($A(\phi)$):
As established in Section 2, a purely conformal transformation preserves null cones. *GW170817 imposes no direct constraint on the magnitude of conformal coupling.* The measurement confirms only that light and gravity share the same causal structure along a single path; it does not constrain the *rate* at which they traverse that path relative to other paths in the universe. The constraint applies only to the *difference* in null cone structures, not the absolute rate of time flow. The symmetry of this exemption must be stated with equal precision: the same null-cone invariance that frees the conformal amplitude from the differential bound also confines its propagation content to the common-mode endpoint calibration — the conformal sector contributes exactly zero propagated

differential delay between any two rays (Section 3.1.3; the static-conformal residual vanishes identically, Paper 19 step_56). The exemption and the inertness are one theorem.

4.3 The Operational Reality: Decoupling the Sectors

To resolve the conflict between the GW170817 constraint and dark matter phenomenology, the two metric sectors must be explicitly distinguished. A frequent objection concerns the Shapiro delay: if TEP posits significant scalar potentials to mimic dark matter, should these not introduce measurable differential delays? The conformal sector can modify the matter-frame clock-transfer calibration associated with a Shapiro-delay measurement, while the geometric null-path delay is carried by the gravitational metric and any backreacted strong-field solution. Same-path EM–GW comparisons cancel common-mode conformal clock effects.

Since Axiom 1 establishes that photons and gravitational waves traverse the same null geodesics defined by the geometric metric, they experience an identical Shapiro delay as they propagate through the gravitational potential. Consequently, a differential arrival-time measurement like GW170817 cancels this common-mode geometric delay entirely. The conformal clock-transfer contribution is also common-mode for same-path EM–GW propagation and cancels in the differential measurement, so the magnitude of the conformal potential is not directly constrained by such differential propagation tests. It remains indirectly constrained by PPN, equivalence-principle, source-screening, gravitational-redshift, and clock-comparison tests — and, more bindingly for this paper's mechanism, by the null-cone invariance itself: the conformal clock-transfer map is a reconstruction-space field (what a lens model would absorb into convergence, mass-sheet normalization, or delay calibration), not a propagating delay. The physically propagating differential residual is the disformal multipath deformation of Section 3.1.4, bounded by this same measurement to $\lesssim 0.2$ s per 2 Mpc halo path.

> **Assumption (Sector Decoupling):** The 10^{-15} bound on the Disformal sector (the "speed of light" vs "speed of gravity") is accepted. The TEP-GL phenomenology relies on Conformal sector gradients (the "rate of time"), which this bound does not directly constrain — and which, by the same null-cone invariance, carry no propagated differential content at all: their observable role is the endpoint/common-mode calibration and reconstruction-space clock-transfer map, bounded by clock-transfer, PPN, redshift, and lensing constraints after environmental screening is applied, with the propagated sector carried separately by the GW170817-bounded disformal channel.

This decoupling makes the TEP framework robust against propagation speed constraints. Consistent with the general disformal relation (Bekenstein 1993), note that the speed of gravitational waves (c_g) constrains the causal structure (the light cone) governed by the disformal term $B(\phi)$, while the chronometric component of Phantom Mass arises from the conformal factor $A(\phi)$ (the clock rate); coherent optical lensing additionally involves scalar backreaction and any permitted disformal response. As long as $c_g = c_\gamma$, the rate at which time accumulates along that path is not determined by EM–GW arrival-time differences. GW170817 strictly constrains cone tilting, but is insensitive to the common-mode conformal time dilation that mimics mass.

Summary of Constraints:

- **Conformal Sector:** Not directly constrained by the GW170817 differential-propagation bound; independently constrained by clock, redshift, PPN, equivalence-principle, source-screening and lensing channels — and, as the binding constraint on the propagation side, identically zero in propagated differential delay by null-cone invariance (Section 3.1.3). Its observable content is the endpoint/common-mode calibration and the reconstruction-space clock-transfer map.
- **Disformal Mean (Monopole):** Tightly constrained by GW170817 ($\lesssim 10^{-15}$).
- **Disformal Gradient (Multipole):** Constrained only by the requirement that the integral of the gradient not exceed the monopole bound excessively. This permits non-trivial millisecond-scale differential structure across the image plane, sufficient for the time-domain signatures predicted in Section 5.

By distinguishing between the *speed of transmission* (directly constrained by GW170817) and the *rate of proper time accumulation* (constrained instead by clock, PPN, and redshift channels, and identically null in propagated differential delay), the conditional phenomenological viability of TEP as a dark-sector reinterpretation is established within the screening and amplitude-closure assumptions stated here — with the propagating discriminator assigned, as in Section 5, to the disformal multipath channel that GW170817 does bound.

4.4 Environmental Screening and Solar System Constraints

A critical quantitative challenge to the TEP framework is the magnitude of the clock-transfer contrast required to affect cluster lensing reconstructions. As established in Box 3.1, halo-scale integration over $L_{\rm halo} \sim 2$ Mpc gives $L_{\rm halo}/c \simeq 6.5 \times 10^6$ yr, so a path-averaged conformal contrast $\langle \Delta A \rangle_{\rm path} \sim 10^{-7}$--$10^{-6}$ yields a year-scale reconstruction-space field — the reconstruction-space quantity of §3.1.3, not a propagated photon delay. This replaces the earlier cosmological-integration estimate, which produced unobserved 10^3--10^5 year gaps and is not the scale required by the Extended Regime. Even at this reduced contrast, if the same field gradients persisted unmodified within the Solar System they would violate high-precision ephemerides constraints (e.g. the Cassini parameter γ).

Screening in TEP is defined at the theory level by the canonical environmental operator, which acts on the observable Temporal Shear $\Sigma_\mu \equiv \nabla_\mu \ln A(\phi)$:

$$\Sigma_\mu^{\rm obs} = \mathcal{S}_\Sigma(\mathcal{E})\,\Sigma_\mu,$$

where the environmental state $\mathcal{E}$ includes source structure, scalar gradients, density, compactness, boundary conditions, and coherence scale. These quantities determine the projection of the underlying temporal field into an observable clock, dynamical, or lensing response. TEP screening is defined directly by the environmental projection of Temporal Shear. No separate screening radius, thin-shell boundary, or imported modified-gravity mechanism is introduced.

Box 4.1: Canonical Temporal-Gradient Suppression

TEP screening acts directly on the observable Temporal Shear,

$$\Sigma_\mu \equiv \nabla_\mu \ln A(\phi),$$

through the canonical environmental operator:

$$\Sigma_\mu^{\rm obs} = \mathcal{S}_\Sigma(\mathcal{E})\,\Sigma_\mu, \qquad 0 \le \mathcal{S}_\Sigma(\mathcal{E}) \le 1.$$

The environmental state $\mathcal{E}$ includes temporal-gradient strength, density, compactness, source structure, boundary conditions, and coherence scale. These quantities determine the projection of the underlying temporal field into an observable clock, dynamical, or lensing response.

In Solar-System conditions,

$$\mathcal{S}_\Sigma(\mathcal{E}_\odot)\,|\Sigma_\mu| \ll |\Sigma_\mu|,$$

suppressing the locally observable response and preserving precision tests of GR. Across extended halo and intergalactic paths,

$$\int_\gamma \mathcal{S}_\Sigma(\mathcal{E})\,\Sigma_\mu\,dx^\mu \neq 0,$$

allowing a cumulative open-path clock-transfer contrast even when the local response is strongly suppressed. The required hierarchy is therefore a hierarchy in the environmental projection of Temporal Shear.

The saturation density ρ_T may be used as an environmental proximity indicator within $\mathcal{E}$, but it is not a hard universal threshold. The fundamental screening object remains $\mathcal{S}_\Sigma(\mathcal{E})$.

5. Observational Predictions: The Era of Chronometric Mapping

The transition from a geometric to a dynamical-time framework shifts the observational focus from static shapes to dynamic arrival times. The "dark sector" is predicted to reveal its true nature not in deep fields, but in high-cadence time-domain surveys. Explicit, falsifiable predictions are presented, organizing them by scale: from the millisecond "jitter" of fast transients (FRBs), to the statistical bias in variable sources, to the global tension between CMB and galaxy lensing.

5.1 The "Jittering" of Lensed Transients

Prediction: Strongly lensed fast transients (FRBs, GRBs) will exhibit achromatic *differential* arrival-time residuals between images that cannot be explained by geometric time delays (Refsdal 1964) or plasma dispersion.

In standard GR, the time delay Δt_{geom} between images is fixed by the mass distribution. In TEP, there is an additional propagated arrival-time residual $\Delta\tilde{\tau}$, carried by the disformal multipath deformation — bounded by GW170817 to $\lesssim 0.2$ s per 2 Mpc halo path — since the conformal sector contributes exactly zero propagated inter-image delay (§3.1.3; Paper 19, step_56). Because ϕ fields in halos may have substructure (or "weather"), this residual varies across the image plane. For a millisecond-duration FRB (e.g., Muñoz et al. 2016), even a tiny gradient in $\Delta\tilde{\tau}$ will manifest as a timing anomaly in the *relative* arrival times of the images. Unlike plasma dispersion (which scales as ν^{-2}), this differential delay is *achromatic* (frequency-independent). The residual is expected to depend on integrated path depth and intervening temporal-field structure, and may therefore correlate with observed redshift; in TEP, redshift here is an observational clock-distance label, not a measure of physical spatial expansion. The anomaly appears as an "Excess Delay vs Redshift" rather than a frequency-dependent sweep. This distinguishes it from "excess Dispersion Measure" (DM), allowing TEP effects to be isolated from plasma effects via multi-frequency observation.

Target Candidates: Recent literature has identified specific anomalies suitable for this test. The repeating source *FRB 20190520B* exhibits a "Dispersion Measure Excess" (~ 900 pc cm^{-3}) relative to its redshift (Koch Ocker et al. 2022), currently attributed to extreme host density. TEP predicts this excess may partially conceal an achromatic temporal delay. Additionally, *FRB 20190308C* (Chang et al. 2024) has been identified as a lensed candidate in the CHIME catalog; any discrepancy between its mass-model time delay and observed delay would constitute direct evidence of the non-geometric temporal shear $\Delta\tilde{\tau}$.

Test Protocol. No securely multiply-imaged FRB with a published independent lens model is available at the time of writing, so no lensed-FRB residual is claimed here as evidence. The prediction is instead stated as an identifiable measurement, using the blind-prediction residual of Axiom 3 (Section 2.3):

$$R_{ij} = \Delta t_{ij}^{\rm obs} - \Delta t_{ij}^{\rm GR},$$

where $\Delta t_{ij}^{\rm GR}$ must be frozen from an independent lens model before the timing comparison is performed. Each element below is required for the test to be diagnostic; a measurement missing any one of them cannot discriminate TEP from astrophysical delay structure.

Required element	Specification
Secure multiple imaging	Consistent sub-arcsecond localization and repeated burst morphology across images
Independent lens model	Frozen, pre-registered prediction $\Delta t_{ij}^{\rm GR}$ with propagated model uncertainty
Observed delay	$\Delta t_{ij}^{\rm obs}$ at high time resolution, corrected for plasma dispersion
Residual	R_{ij}, with plasma, microlensing, and host-scattering budgets subtracted (Section 2.3)
Discriminator	Achromatic residual with non-zero ensemble mean, correlated with path environment

Table 5.1: Required elements of a diagnostic lensed-FRB timing test. The achromaticity requirement separates a temporal-transport residual from plasma dispersion ($\propto \nu^{-2}$); the non-zero ensemble mean separates it from microlensing, which averages to zero with RMS scaling as $1/\sqrt{N}$.

5.2 The Variability Scatter Relation

Prediction: The *dispersion* (scatter) of weak-lensing shear measurements should correlate with the variability/kinematics of the background source population.

Whereas on the isotropic-conformal benchmark ($\tilde{g}_{\mu\nu} = A^2 g_{\mu\nu}$) scalar backreaction and spatial-disformal configurations fall short of a CDM-free coherent halo-lensing amplitude, under the canonical unstretched-space temporal metric ($\tilde{\gamma}_{ij} = \gamma_{ij}$, §6.9 Step 02; Paper 18 Step 21; Paper 19 Step 64) intermediate temporal depth $-\beta_A u_{\rm grav}(\mathbf{x})$ on the selected two-branch action supplies the coherent static convergence and Fermat potential directly from baryons alone. The source-evolution response is a distinct, bounded reconstruction component. The secondary "stochastic shear" term depends on source proper motion $\vec{\mu}_s$. Because $\vec{\mu}_s$ is randomly oriented, this term adds a random vector to the shear signal. TEP predicts that if one constructs a shear map using highly variable or fast-moving sources, the shear RMS will be systematically higher than for static sources, even if the mean profile (the halo) is identical — an excess over the standard-map baseline, which already predicts the same ordering. The predicted excess is bounded by the Box 3.2 ledger values ($\gamma^{\rm TC} \lesssim 6 \times 10^{-7}$ at the reconstruction-space scale, $\sim 10^{-15}$ on the propagating carrier), so the test operates as a precision bound on the envelope amplitude at whatever scatter precision is achieved (Box 5.1). Because the ordinary Fermat surface generates the same correlation at $\gamma^{\rm std} \sim 3 \times 10^{-8}$ for strong-lens geometries ($\sim 6 \times 10^{-10}$ galaxy–galaxy; step_01), the discriminating content is the dispersion residual over that standard-map baseline, not the correlation's existence.

Box 5.2: The Einstein Cross Test (Existing Data)

The Einstein Cross (Q2237+0305) provides a unique opportunity to test TEP with *existing* archival data. This quadruply-imaged quasar has:

- **High variability:** The background quasar shows significant optical variability on timescales of weeks to months.
- **Anomalous flux ratios:** The observed image flux ratios deviate from smooth lens model predictions, typically attributed to microlensing by stars in the lens galaxy.
- **Extensive monitoring:** Decades of photometric data exist (OGLE, Gaia, HST).

Amplitude estimate. Two response routes connect the temporal field to the observed flux ratios, and the bookkeeping of Box 3.2 applies to both. The Jacobian route gives $\delta\mu/\mu \sim \mu\,\gamma^{\rm TC}$: at magnification $\mu \sim 10$ this is $\sim 6 \times 10^{-6}$ at the reconstruction-space scale and $\sim 10^{-14}$ on the propagating carrier — four orders of magnitude below the observed 10–30% anomalies even at the envelope. The emission-time route gives $\delta F/F \sim \dot{\nu}\,\delta t_{\rm em}$: for a quasar varying fractionally at $\dot{\nu} \sim 10\%$ per month, the propagating-carrier delay ($\lesssim 0.2$ s) contributes $\delta F/F \sim 10^{-8}$, while a year-scale propagating reading of the clock-transfer map would produce order-unity phase-correlated flux swings that are not observed. The Einstein Cross anomalies therefore remain in the stellar-microlensing regime and are not claimed as a TEP signal at any admissible amplitude.

TEP Prediction: The discriminating content of this test is the correlation structure, not the anomaly amplitude. If a residual emission-time contrast $\delta t_{\rm em}$ exists between the four sightlines, the apparent flux-ratio anomaly amplitude scales with the source's instantaneous variability rate:

- During periods of rapid quasar variability, the temporal-composite contribution to flux ratios is *larger* ($\delta F/F \propto \dot{\nu}\,\delta t_{\rm em}$).
- During quiescent periods, it regresses toward the smooth-lens prediction.
- Unlike stellar microlensing, which is uncorrelated with the source's intrinsic state, the temporal contribution is coherent with the quasar's own variability phases—effectively turning "on" during violent source activity.

Status: This test can be performed immediately using archival OGLE light curves cross-correlated with flux ratio measurements, and it functions quantitatively as a bound: at $\sim 1\%$ photometric precision and $\dot{\nu} \sim 10\%$ per month, the absence of a variability-phase-correlated residual bounds any coherent emission-time contrast at $\delta t_{\rm em} \lesssim$ a few days — sitting between the millisecond propagating carrier and the excluded year-scale propagating reading. Detection of the conservative-baseline carrier itself requires the arrival-time residual protocol of §5.1. This is flagged as a priority archival analysis.

5.3 The CMB-Galaxy Lensing Tension

Prediction (Regime II): TEP-GL predicts that galaxy weak-lensing measurements can contain a source-dependent covariance contribution absent from CMB lensing, even when their mean inferred amplitudes are statistically consistent.

While the CMB is effectively a static backlight (zero intrinsic evolution), galaxy sources are dynamic population. Under TEP, the stochastic component of the clock-transfer signal (driven by source proper motion, see Section 5.2) introduces an additional "kinematic noise" to galaxy shear measurements. Because the CMB is static, it is immune to this effect. Consequently, TEP predicts that precision cosmology inferred from galaxy weak lensing carries an unmodeled source-dependent covariance term absent from CMB lensing. The quantitative reach of that term is set by Box 3.2: its variance contribution $\gamma^2_{\rm TC} \sim 4 \times 10^{-13}$ at the reconstruction-space scale ($\sim 2 \times 10^{-30}$ on the propagating carrier) is far below the shape-noise variance ~ 0.09, so it cannot supply the percent-level amplitude of the observed S_8 tension. The prediction is the existence and source-class structure of the covariance term, not an explanation of the tension's magnitude.

Box 5.3: Source-Class-Dependent Shear Covariance

Observed Situation: Current weak-lensing results are survey-dependent: DES Y6 continues to favour a lower S_8 than the combined CMB determination, whereas KiDS-Legacy is statistically consistent with Planck. TEP-GL therefore does not identify a universal mean S_8 offset as its prediction; its distinctive prediction is an additional source-dependent covariance that should track source temporal structure and survey selection.

- *Planck* CMB (2018): $S_8 = 0.834 \pm 0.016$
- DES Y6 (2025): $S_8 = 0.789 \pm 0.012$, $\sim 2.6\sigma$ from combined Planck+ACT+SPT
- KiDS-Legacy (2025): $S_8 = 0.815^{+0.016}_{-0.021}$, 0.73σ from Planck

TEP Interpretation: Previous iterations of TEP considered secular source evolution as a driver for coherent shear offsets, but the magnitude ($\sim 10^{-6}$) is too small to explain the 5% tension. The stochastic-shear covariance term is likewise bounded: $\gamma^2_{\rm TC}/\sigma^2_{\rm sh} \sim 4 \times 10^{-12}$ relative to shape noise at the reconstruction-space scale, so this channel does not resolve the S_8 tension; its role is diagnostic — a source-dependent covariance signature whose presence, not its magnitude, is the observable.

Mechanism: Standard maximum-likelihood estimators weight data by the inverse covariance (C^{-1}). Standard analyses assume shear noise is dominated by random galaxy orientations ("shape noise"). If the covariance matrix omits a source-dependent kinematic term (σ^2_μ), the relative weighting of high-variance regions such as cluster outskirts is misspecified, and the recovered clustering amplitude is biased.

- **Source-Dependent:** Correlated with galaxy type and redshift (via proper motion).
- **Unmodeled:** Not present in standard covariance matrices.

Result: Unmodeled excess variance in the likelihood analysis acts as a potential source of systematic bias. However, a zero-mean covariance term does not by itself determine the sign of the inferred S_8 shift; establishing the direction requires an explicit estimator-level calculation for each survey pipeline, which is not carried out here. The CMB, being static, carries no such kinematic covariance term.

Prediction: The falsifiable TEP-GL prediction is not a specific S_8 shift but the existence of an additional source-dependent shear covariance: after standard shape-noise controls, residual shear covariance should vary with source variability, proper motion, or temporal structure. The cosmological mean-growth prediction belongs to the TEP-HC (Paper 18) perturbation closure; the GL-specific discriminator is this source-class dependence. Whether it contributes to the observed S_8 tension, and in which direction, depends on the survey estimator.

5.4 Comparison of Predictions: TEP-GL vs. Particle Dark Matter

The following table summarizes the distinguishing predictions of the two frameworks:

Observable	Particle DM Prediction	TEP-GL Prediction
Lensed FRB timing	GR lens delays from baryons, particle dark matter and substructure; no TEP-specific residual correlated with temporal-field environment after lens-model controls	Achromatic *residual* anomaly (ms-scale)
Source-dependent Shear	No TEP-specific source-temporal correlation after intrinsic-alignment, selection and measurement-systematic controls	**Excess Scatter** (noise scales with μ_s; bounded amplitude per Box 3.2)
Source variability–lens mass correlation	None (mass is source-independent)	Source-class-dependent reconstruction residual correlated with source temporal structure; amplitude determined by $\kappa_{\rm lens}$ and the environmental transfer function
CMB lensing	Standard convergence	Source-dependent temporal-composite term absent at leading order; under the unstretched-space temporal metric on the selected two-branch action, intermediate temporal depth boosts baryon-only cosmic-web $C_\ell^{\phi\phi}$ by $9.4\times$–$129\times$, recovering the Planck lensing amplitude without CDM (Paper 18, Step 21)
Galaxy weak lensing	Source-independent shear consistent with the physical mass distribution; no TEP-specific source-temporal covariance	Isothermal-halo convergence $\kappa(b) \propto 1/b$ from unstretched-space temporal depth on the low-gradient $k\sqrt{\xi}$ branch, plus bounded source-class-dependent temporal-composite covariance for evolving sources

Observable	Particle DM Prediction	TEP-GL Prediction
Chromatic dependence	None	None (both achromatic)
Direct detection experiments	Expected signal (WIMP recoil accounting for lensing/rotation/CMB)	No particulate signal; dynamical Temporal Shear and unstretched-space temporal refraction account for rotation curves, wide binaries, cluster lensing, and CMB lensing from baryons and the selected two-branch scalar action

A distinctive TEP-GL discriminator is the additional source-dependent component of the inferred lensing signal above the standard delay-map baseline. The coherent scalar-metric contribution is source-independent, while the temporal-composite contribution predicts a residual correlated with source variability, proper motion, or temporal structure beyond what the ordinary Fermat surface supplies ($\gamma^{\rm std}$, step_01).

5.5 Falsification Criteria

TEP-GL is a falsifiable hypothesis. The following observations would exclude specific regimes or the framework entirely:

1. **conservative baseline Exclusion:** If precision timing of strongly lensed FRBs yields achromatic residuals consistent with zero to better than 0.1 ms across diverse lens environments, the conservative baseline parameter space is excluded.
2. **Universal Shear Scatter:** If the intrinsic scatter of weak-lensing shear measurements, after correcting for measurement noise, equals the standard delay-map prediction $\gamma^{\rm std} = \mu_s\, \partial_j \Delta T_{\rm Fermat}$ across all source kinematic classes — i.e., no residual above the baseline — the stochastic temporal shear mechanism is excluded at that precision.
3. **CMB-Galaxy Agreement:** CMB–galaxy agreement below 1%, after source and survey controls, excludes a temporal-composite contribution above that level. CMB lensing directly constrains the physical deflection potential, independent of galaxy-source variability. A CDM-free TEP model that cannot reproduce this potential fails the proposed dark-sector replacement, even if its dynamical phantom-mass channel survives as a separate response. Persistent lensing–dynamics agreement in low-acceleration systems also bounds the shear share of the dynamical anomaly.
4. **Phantom Mass (decisive):** In isolated, low-acceleration systems where $S_\Sigma \to 1$, the mass inferred from dynamics exceeds the single-pass weak-lensing mass — by the full shear share on the isotropic-conformal benchmark ($g_{\rm lens}^{\rm iso} \simeq g_{\rm bar}$), and by the factor-of-two scalar-potential weight under the canonical unstretched-space metric ($\Phi_{\rm dyn} = \Phi_N + c^2\beta_A u_{\rm grav}$ vs $\Phi_{\rm lens} = \Phi_N + \tfrac{1}{2}c^2\beta_A u_{\rm grav}$, so $M_{\rm lens}^{\rm u.s.}/M_{\rm dyn} = (1+\beta_A^2 y_{\rm eff})/(1+2\beta_A^2 y_{\rm eff})$; Paper 0 Eq. 63). Environment- and acceleration-resolved comparison of rotation-curve and weak-lensing mass on the same isolated systems is a sharp dark-sector test: detection of a systematic signed dynamical–lensing difference at matched radii after marginalizing over anisotropy, projection, and external-convergence uncertainties—the differential phantom mass—is a distinctive positive signature not generically expected from a single shared collisionless mass distribution; its persistent absence on isolated single-pass halos constrains the single-pass versus multi-component weighting of the unstretched-space optical response (§6.9).
5. **Chromatic Anomaly:** If any timing or morphological anomaly shows wavelength dependence after correction for plasma dispersion and dust, the achromatic prediction of two-metric coupling is falsified. This would indicate conventional astrophysical systematics rather than metric effects.
6. **Direct Detection:** A confirmed detection of dark matter particles that quantitatively accounts for the relevant lensing, rotation-curve, and CMB phenomenology would falsify the strong TEP dark-sector ontology. Detection of an exotic particle that does not account for these observations would not by itself falsify TEP; the particle would be an additional component, and TEP would then be tested by whether temporal-transport geometry still predicts residuals unaccounted for by the particle model.

Joint satisfaction of these null thresholds excludes the distinctive chronometric and temporal-composite predictions at the amplitudes specified here. The coherent scalar-backreaction sector is separately tested by CMB and galaxy lensing consistency against its frozen HC/GL prediction. The framework does not claim immunity from observation; the claim is that the observations required to test TEP have not yet been performed with sufficient precision.

Box 5.1: Quantitative Falsification Thresholds

To ensure rigorous falsifiability, explicit null-result thresholds are specified for each prediction channel:

Test	Sample Size	Precision Required	Null-Result Threshold	Regime Falsified
Lensed FRB Timing	≥10 lensed FRBs	<0.1 ms timing	95% upper bound on ensemble achromatic residual amplitude below 0.1 ms	conservative baseline
Shear-Variability Correlation	≥1000 lenses per class	<0.5% shear scatter	No correlation at >3σ	temporal-composite dynamic channel
CMB-Galaxy S_8 Tension	CMB-S4 + LSST	<1% S_8 agreement	Statistically consistent agreement (< 1σ)	temporal-composite contribution above 1%
Variability-Mass Correlation	≥100 lenses with varied sources	10% mass precision	No correlation at >3σ	Source-dependent reconstruction channel

Amplitude probed by each threshold. The thresholds act on different carriers (Box 3.2), and the table is populated accordingly. The lensed-FRB row acts on the propagating delay field directly: a 0.1 ms ensemble bound excludes the disformal multipath carrier at three orders of magnitude below the GW170817-saturating level (~ 0.2 s per 2 Mpc halo path) — equivalently, it bounds the disformal normalization $D(\partial_{\hat{n}}\phi)^2$ to $\sim 2 \times 10^{-18}$ on halo paths, the flagship, fully powered test of the same $B(\phi)$ instrument the holonomy program probes on the odd-parity channel. The shear-variability row bounds $\gamma^{\rm TC} \lesssim 5 \times 10^{-3}$, equivalent to a delay-gradient bound $\nabla(\Delta\tilde{\tau}) \lesssim 2 \times 10^4$ yr/arcsec — a weak upper limit on the reconstruction-space scale, not a test of the carrier. The S_8 row bounds fractional source-dependent covariance at the percent level; the temporal-composite term sits at $\sim 4 \times 10^{-12}$ relative to shape noise even at the envelope, so this row's content is the structural existence test rather than an amplitude reach. The variability–mass row probes $\delta\kappa \sim \gamma^{\rm TC} \gtrsim 10^{-1}$ directly, and through the delay-absorption route bounds any propagating residual at $\lesssim 10\%$ of the Fermat delay (~ 10 d per 100 d) — already excluding a propagating year-scale reading, consistent with the conformal null of §3.1.1. Together the rows bound the propagating delay field at every scale between the millisecond and the tens-of-days level, with the chronometric sector confined to the reconstruction space where the Paper-19 residual program operates.

Joint Channel Exclusion: If all four null-result thresholds are met simultaneously, the distinctive chronometric and temporal-composite predictions of TEP-GL are excluded at the amplitudes specified here. The coherent scalar-backreaction sector remains independently testable through its frozen HC/GL prediction for CMB and galaxy lensing. TEP-GL as a complete lensing realization is excluded only if both the time-domain/source-dependent channels and the independently specified coherent optical prediction are ruled out.

Timeline: Tests (1) and (4) are achievable within 5–10 years (CHIME, DSA-2000, Rubin/LSST). Tests (2) and (3) require next-generation surveys (CMB-S4, Euclid) with ~10-year horizons. A definitive verdict on TEP-GL is therefore expected by ~2035.

6. Discussion: The Dark Sector — Phantom Mass in Dynamics and Lensing

6.1 Limitations of the "Stack of Corrections"

Gravitational physics has historically advanced by refining the fundamental geometric framework rather than adding ad-hoc corrections. TEP represents a similar upgrade to the fundamental object, following the natural historical progression of physical theory:

- **Newton:** Gravity is a Force; Time is Absolute.
- **Einstein:** Gravity is Geometry; Time is Relative (Coordinate-Dependent).
- **TEP:** Gravity is Geometry; Time is a *Dynamical Field*.

By treating the rate of proper time accumulation as a physical field with its own degrees of freedom (rather than a fixed function of the metric), TEP unifies the "stack" into a single framework. The "dark matter" anomaly is the observation that this field has spatial gradients.

6.2 CMB Lensing and the Integrated Sachs-Wolfe Constraint

A scalar field capable of generating year-scale reconstruction-space contrasts on cluster scales raises a critical question: what are the implications for the Cosmic Microwave Background (CMB)? Two potential tensions must be addressed.

6.2.1 The ISW Effect

The Integrated Sachs-Wolfe (ISW) effect arises when CMB photons traverse time-evolving gravitational potentials. In TEP, the conformal factor $A(\phi)$ contributes an additional term to the photon temperature perturbation:

$$\left.\frac{\Delta T}{T}\right|_{\rm TEP} = \int \frac{\partial \ln A(\phi)}{\partial t}\, dt$$

For TEP to remain consistent with *Planck* constraints on the ISW amplitude, one of the following must hold:

- **Quasi-Static Regime:** The scalar field evolves slowly ($\partial_t A \ll H_0 A$), so the TEP contribution to ISW is subdominant to the standard $\dot{\Psi}$ term. This is natural if ϕ tracks the matter distribution adiabatically.
- **Cancellation:** In some scalar-tensor theories, the conformal ISW contribution partially cancels the metric ISW, leaving the net effect within observational bounds (cf. Amendola et al. 2008).

Box 6.1: Order-of-Magnitude ISW Consistency Check

Conditional on the HC tracking closure in which ϕ follows the matter distribution adiabatically, the resulting order-of-magnitude estimate satisfies the quasi-static ISW bound.

Setup: The conformal factor is $A(\phi) = e^{-u} \approx 1 - u$, with the field excursion $u \sim 10^{-6}$ on halo scales (u is the canonical dimensionless field of Paper 0; the symbol $\alpha(\phi) \equiv d\ln A/d\phi = \beta_A$ is reserved for the coupling function). The following estimate uses the conventional reference-background variables employed by the HC linear-transfer representation (Paper 18). Here $H_{\rm ref}(z)$ is an observational/reference inverse-time scale and $a_{\rm eff} = A_{\rm clock}$; neither denotes physical expansion of the underlying spatial geometry. The ISW constraint requires:

$$\left|\frac{\partial \ln A}{\partial t}\right| = |\dot{u}| \ll H_0$$

Estimate of $\dot{u}$: If ϕ tracks the matter distribution adiabatically (sourced by T^{μ}_{μ}), then u evolves on the timescale of structure formation:

$$\dot{u} \sim u \cdot \frac{\dot{\rho}}{\rho} \sim u \cdot H_{\rm ref}(z) \cdot f_{\rm ref}(z)$$

where $f_{\rm ref}(z) \equiv d\ln D/d\ln a_{\rm eff} \approx \Omega_m(z)^{0.55}$ is the growth rate and $a_{\rm eff} = A_{\rm clock}$. At $z \sim 0.5$ (peak ISW sensitivity), $f_{\rm ref} \approx 0.8$ and $H_{\rm ref}(z) \approx 1.2 H_0$.

Result:

$$\frac{|\dot{u}|}{H_0} \sim u \cdot f_{\rm ref} \cdot \frac{H_{\rm ref}(z)}{H_0} \sim 10^{-6} \times 0.8 \times 1.2 \approx 10^{-6}$$

This is six orders of magnitude below the ISW constraint threshold ($\lesssim 0.1$). The quasi-static approximation is therefore satisfied under this tracking closure; the result should not be interpreted as independent derivation of that closure within TEP-GL.

Physical Interpretation: The scalar field ϕ varies spatially (creating "dark matter" halos) but evolves temporally only as fast as the underlying matter distribution. Year-scale *spatial* delays across an image plane are compatible with cosmologically slow *temporal* evolution because the delay gradient is set by the static ∇u, not by $\dot{u}$.

6.2.2 CMB Lensing Power Spectrum

The CMB lensing convergence power spectrum $C_\ell^{\kappa\kappa}$ is measured to high precision by *Planck* and ACT. In standard ΛCDM, this is sourced entirely by the integrated mass distribution. Consistent with the optical-tidal decomposition of §3.1.4, the physical angular

remapping in TEP is carried by the baryonic term, the scalar-backreaction correction to the Einstein-frame geometry, and any permitted disformal contribution:

$$\kappa_{\rm TEP}^{\rm phys} = \kappa_{\rm baryon} + \delta\kappa_{g[\phi]} + \delta\kappa_B$$

The conformal sector may additionally enter the *inferred* convergence through the chronometric reconstruction mapping,

$$\kappa_{\rm TEP}^{\rm inferred} = \kappa_{\rm TEP}^{\rm phys} + \delta\kappa_{A,\,\rm recon},$$

on the isotropic-conformal benchmark ($\tilde{g}_{\mu\nu} = A^2(\phi) g_{\mu\nu}$, where isotropic conformal rescaling preserves null cones, Axiom 1), whereas under the canonical unstretched-space temporal metric (§6.9) the local lapse $A_{\rm loc}(u_{\rm grav})$ refracts the CMB rays directly across unstretched space.

For the CMB (a static, non-evolving source), the temporal-composite response term vanishes identically—there is no source proper motion or intrinsic variability on lensing timescales to couple to the delay gradient. Removing that term does *not* make the CMB insensitive to TEP: it means CMB lensing constrains the coherent scalar-metric perturbation sector rather than source-evolution smearing.

Implication: CMB lensing tests the source-independent optical potential, not the temporal-composite response of a variable galaxy. While the baseline `hi_class` implementation in Paper 18 uses isotropic conformal rescaling $\tilde{g}_{\mu\nu} = A^2(\phi) g_{\mu\nu}$ (which cancels out of null rays and therefore retains a standard CDM fluid $\Omega_{\rm cdm} h^2 = 0.1133 \pm 0.0061$ in the active-perturbation posterior, 0.1148 ± 0.0042 in the full background+lensing chain, leaving 8.72% of $C_{100}^{\phi\phi}$ when `omega_cdm` is removed), Paper 18 Step 21 demonstrates that under the unstretched-space temporal metric $\tilde{g}_{\mu\nu} = -A_{\rm loc}^2(u_{\rm grav}(\mathbf{x}))U_\mu U_\nu + A_{\rm sp}^2(u_{\rm drift}(t,\mathbf{x}))\Pi_{\mu\nu}$ ($U_\mu \equiv -\nabla_\mu\bar{\phi}/\sqrt{2\bar{X}}$) on the selected two-branch action $P_{,X} = k\sqrt{\xi} + 2\xi$ ($k = 16.03$), intermediate temporal depth along the photon path amplifies baryon-only cosmic-web and cluster-outskirt potentials by $M = 3.07$–11.37, boosting $C_\ell^{\phi\phi}$ by $9.4\times$–$129\times$ and recovering the Planck lensing power scale without a CDM fluid.

6.2.3 Parameter Space Constraints

Combining ISW and CMB lensing constraints with the continuous gradient-suppression screening framework (Box 4.1) defines a narrow but viable parameter window:

Constraint	Requirement	Status
Solar System (Cassini)	$\mathcal{S}_\Sigma(\mathcal{E}_\odot)\,\lvert\Sigma\rvert \ll \lvert\Sigma\rvert$	Satisfied via quadratic-branch kinetic screening ($y \sim (r/r_*)^{4/3} \ll 10^{-5}$; Paper 0, Step 57)
Cluster Lensing	Optical-sector requirement: $\kappa \sim 10^{-1}$ at cluster arcs and subcluster galaxies	**Closed under unstretched-space temporal refraction:** while isotropic-conformal stress-energy is Planck-suppressed ($\kappa_\phi \simeq 6.5 \times 10^{-6}$; Paper 19, Step 64) and spatial-disformal $B(\partial\phi)^2$ is bounded $\sim 5\times$–$70\times$ short, the unstretched-space temporal metric on the selected two-branch action reproduces the SN Refsdal Fermat delay ($+330.9$ d sheet, $+221.6$–443.2 d 3-D ray; Paper 19, Step 64) and the Bullet Cluster galaxy-peak convergence ($\kappa_{\rm tot} = 0.307$–0.372, no moat; §6.9, Step 02) from baryons alone
ISW (*Planck*)	$\partial_t \ln A/H_0 \lesssim 0.1$	Satisfied on the kinetically screened $w_c \gtrsim 700$ closure (Paper 18)
CMB Lensing	Source-independent lensing spectrum $C_\ell^{\phi\phi}$ from baryons and the temporal field	**Closed at the cosmic-web scale (Paper 18, Step 21):** unstretched-space temporal refraction on the selected two-branch action boosts baryon-only $C_\ell^{\phi\phi}$ by $9.4\times$–$129\times$ across $R = 1$–10 Mpc filaments and cluster outskirts, recovering the Planck lensing amplitude without CDM

The comparison across channels is therefore sharp: under isotropic conformal rescaling $\tilde{g}_{\mu\nu} = A^2(\phi) g_{\mu\nu}$, spatial stretching cancels $A^2(\phi(x))$ out of null rays and leaves only the Planck-suppressed scalar stress tensor ($\kappa_\phi \simeq 6.5 \times 10^{-6}$ on the selected sector; Paper 19, Step 64) and the bounded disformal cone tilt (Step 63). Under the unstretched-space temporal metric $\tilde{g}_{\mu\nu} = -A_{\rm loc}^2(u_{\rm grav}(\mathbf{x}))U_\mu U_\nu + A_{\rm sp}^2(u_{\rm drift}(t,\mathbf{x}))\Pi_{\mu\nu}$ ($U_\mu \equiv -\nabla_\mu\bar{\phi}/\sqrt{2\bar{X}}$), intermediate clock rates $A(\phi(x))$ along the ray do not cancel, supplying the baryon-sourced cluster, strong-lensing, and CMB lensing convergence from the same two-branch action $P_{,X} = k\sqrt{\xi} + 2\xi$ ($k = 16.03$ fixed by a_0, zero per-system free parameters) that governs galactic rotation curves and wide binaries. Unlike superfluid dark matter (Berezhiani & Khoury 2015), which postulates a new eV-scale dark boson species that clusters as dust on linear scales and undergoes a thermal BEC transition into a $P(X) \sim X^{3/2}$ phonon superfluid inside cold galaxies (breaking down in hot galaxy clusters where $T > T_c$), TEP posits zero dark-matter particles ($\omega_{\rm cdm} = 0$): across a rich X-COP galaxy cluster (

$M_b(< R_{200}) = 1.45 \times 10^{14}\,M_\odot$, $R_{200} = 2.0$ Mpc, $R_{500} = 1.2$ Mpc; Ghirardini et al. 2019; Eckert et al. 2019), the extended ICM profile ($\beta \simeq 0.65$) keeps $q(R) = g_b(R)/g_t = 0.015$--$0.040 \ll 1$ across $R = 0.5$--2.0 Mpc so the cluster ICM never enters the $q \gtrsim 1$ high-acceleration screened regime of galaxy disks. Under the exact static two-branch Gauss law $(k/\sqrt{2})q(r)y(r)^2 + q(r)^2 y(r)^3 = 1$ on the empirical X-COP baryon profile (replacing the earlier $\beta = 1$ Plummer benchmark that placed 72% of cluster baryons inside 0.5 Mpc and gave $5.78, 4.61, 3.47, 2.39$), the local 3D ratio $1 + 2\beta_A^2 y_{\rm tb}(R)$ is $5.80, 5.16, 4.80, 4.59, 3.98$ across $R = 2.0, 1.5, 1.2, 1.0, 0.5$ Mpc with integrated static potential-depth ratio $\Phi_{\rm tot}(R)/\Phi_b(R) = 7.64, 6.85, 6.44, 6.20, 5.61$; when the pre-boundary clock-slope well ($\omega_\phi/\omega_b = y_s/\sqrt{2} = 5.26$, $\sigma_{T,p} = 0$) retains its unshocked NFW collapse concentration ($c_{200} = 4.0$) while virial shock heating and AGN feedback deplete collisional ICM gas in the core ($f_{\rm ret}(R) = f_b(< R)/f_b(R_{200})$ dropping from 1.00 at 2.0 Mpc to 0.90 at 1.2 Mpc and 0.68 at 0.5 Mpc; Eckert et al. 2019), the virialized enclosed ratio $M_{\rm dyn}(< R)/M_{b,\rm obs}(< R) = 1 + (y_s/\sqrt{2})/[f_{\rm ret}(R)(1 + q(R))]$ rises inward from 6.18 at $R = 2.0$ Mpc ($M_{\rm dyn} = 8.96 \times 10^{14}\,M_\odot$) and 6.41 at $R = 1.5$ Mpc ($6.96 \times 10^{14}\,M_\odot$) through 6.73 at $R = 1.2$ Mpc ($5.59 \times 10^{14}\,M_\odot$) and 7.12 at $R = 1.0$ Mpc ($4.61 \times 10^{14}\,M_\odot$) to 8.44 at $R = 0.5$ Mpc ($1.99 \times 10^{14}\,M_\odot$; Paper 18, Step 21 Part D), matching observed X-COP hydrostatic and weak-lensing profiles ($\simeq 6$ at R_{200}, 6--7 at R_{500}, and 8--10 at 0.5 Mpc) without a MOND cluster deficit, thermal condensate failure, or collisionless NFW particle cusp. The baryon-retention profile $f_{\rm ret}(R)$ is an observational input from X-COP, so the inner-radius agreement tests the unshocked, collisionless distribution of the clock-slope mode given the observed gas depletion; the zero-parameter prediction is the normalization $1 + y_s/\sqrt{2} \simeq 6.2$ at R_{200}.

Box 6.3: Cross-Scale Temporal-Gradient Constraint

A viable TEP realization must produce a strongly suppressed observable Temporal Shear in local high-precision environments while retaining a nonzero integrated temporal-gradient response across extended halo paths:

$$\mathcal{S}_\Sigma(\mathcal{E}_\odot)\,|\Sigma| \leq \Sigma_{\rm local}^{\rm max},$$

$$\left| \int_{\gamma_{\rm halo}} \mathcal{S}_\Sigma(\mathcal{E})\,\Sigma_\mu\,dx^\mu \right| \sim 10^{-7}\text{--}10^{-6}.$$

These are constraints on one continuous environmental operator, not independently adjusted couplings at different scales. The Solar-System bound is satisfied when the environmental projection $\mathcal{S}_\Sigma(\mathcal{E}_\odot)$ suppresses the locally observable Temporal Shear below precision-test thresholds. The halo constraint is satisfied when the same operator, acting across an extended path through lower-density environments, permits a cumulative clock-transfer contrast of order 10^{-7}--10^{-6} on the matter-clock congruence — the reconstruction-space field of §3.1.3, which is endpoint-determined for photons and is not a propagated delay.

6.3 Interpretational Challenges

TEP effects mimic standard lensing signatures, making them difficult to distinguish without time-domain analysis.

1. **Static Degeneracy:** Most lensing data are analyzed under static assumptions. In this limit, the conformal sector enters as a reconstruction degeneracy — the endpoint-determined clock-transfer map defined through $A(\phi) = \exp(\beta_A \phi/M_{\rm Pl})$ and the line-of-sight field profile $\phi(r)$ is absorbed into mass-sheet and delay normalization rather than into propagated arrival times (Section 3), while optical convergence and shear are carried by the scalar-backreaction and disformal channels. The predicted reconstruction-space map is qualitatively similar to the observed map; a full quantitative comparison requires the source light-curve model, which is the next pipeline step.
2. **Differential vs. Common-Mode Effects:** The GW170817 constraint is often interpreted as excluding modified metric couplings. However, as shown in Section 4, this constraint applies to the differential disformal sector. Conformal proper-time dilation is a *common-mode effect* for co-propagating signals, leaving the "speed of gravity" constraint satisfied while permitting significant time-domain phenomenology.

6.4 Reinterpreting the Evidence: Multipath vs. Single-Path

When viewed through the lens of TEP, the disparate constraints on the dark sector resolve into a consistent picture based on signal topology:

- **Lensing (Multipath):** Observations that require "dark matter" (e.g., lensing) are fundamentally *multipath* measurements. These different pathways sample different values of the time field ϕ, creating a differential clock-transfer field that isochronous

reconstruction reads as "phantom mass," while any propagated differential delay rides only on the backreacted and disformal sectors.
- **GW170817 (Single-Path):** Observations that constrain "modified gravity" (GW170817) are fundamentally *single-path* measurements. The signals originate from the same coordinate and traverse the same time-warped regions. The temporal distortions are common-mode and cancel out.

The apparent contradiction—"how can gravity be modified enough to create dark matter but unmodified enough to pass GW170817?"—is resolved. TEP introduces the temporal field as the additional physical degree of freedom; its conformal sector governs clock transport without bending null geodesics or shifting propagated arrival times, while its stress-energy and strong-field operators can backreact on the Einstein-frame geometry (Bahrain, Paper 28). The differential-propagation bound constrains the disformal sector, not the common-mode conformal clock sector. Time manifests as a propagated differential observable only through non-exact transport on divergent paths; in the exact conformal sector it manifests as a reconstruction degeneracy.

6.5 Path-Dependent Distance/Proper-Time Ratio

If TEP is correct, then local c remains invariant, but the *inferred ratio between spatial separation and matter-clock transfer registered between endpoints* along extended paths is path dependent. This challenges the foundational assumption used to interpret all astrophysical data.

- **Standard Assumption:** The universal clock rate is constant; therefore, any anomaly in t must be due to extra path length (spatial curvature) or extra mass (Shapiro delay). Conclusion: particulate *dark matter* is inferred.
- **TEP Reality:** The matter proper-time accumulation rate $d\tilde{\tau} = A(\phi)d\tau_g$ varies with location. The anomaly in inferred travel time is due to the scalar field gradient modulating the clock-transfer field registered by matter congruences along the path, not to a variation in local c. Conclusion: apparent *dark matter* is reconstructed temporal transport.

TEP removes the need for an invisible substance by correcting the assumption that a single universal clock rate governs all paths, rather than by asserting a variable speed of light.

This is the local multipath analogue of the cosmological TEP mapping: conventional cosmology reconstructs observed redshift as spatial expansion, whereas TEP attributes it to the evolution of the conformal proper-time map on an underlying static spatial geometry (Athens, Paper 26; Thika, Paper 27). In lensing, the corresponding reconstruction can project temporal transport into inferred spatial mass. The two effects — dark matter and cosmic expansion — are parallel temporal-to-spatial reconstruction artefacts in TEP, not independent ad hoc mechanisms.

6.6 TEP as a Bridge Between Paradigms

TEP offers a potential unification of competing frameworks:

- **CDM:** For the dynamical sector, TEP offers a candidate temporal-field origin for the mass anomaly ordinarily ascribed to Cold Dark Matter: the Temporal Shear acts on matter trajectories as the extra pull Newtonian analysis counts as a halo. In the lensing sector, under the historical isotropic-conformal benchmark ($\tilde{g}_{\mu\nu} = A^2 g_{\mu\nu}$) $A^2(\phi)$ cancels out of null rays so real lensing is baryon-sourced, whereas under the canonical unstretched-space temporal metric ($\tilde{\gamma}_{ij} = \gamma_{ij}$, §6.9 Step 02) intermediate temporal depth $-\beta_A u_{\rm grav}(\mathbf{x})$ refracts photons directly along the line of sight ($\Phi_{\rm lens} = \Phi_N + \frac{1}{2}c^2\beta_A u_{\rm grav}$), reproducing the Bullet Cluster galaxy-peak convergence ($\kappa_{\rm tot} = 0.307$–0.372), the SN Refsdal Fermat delay (Paper 19, Step 64), and Planck CMB lensing (Paper 18, Step 21) from baryons and the selected two-branch action $P_{,X} = k\sqrt{\xi} + 2\xi$ while predicting a testable single-halo dynamical–lensing ratio $M^{\rm u.s.}_{\rm lens}/M_{\rm dyn} = (1+\beta_A^2 y_{\rm eff})/(1+2\beta_A^2 y_{\rm eff})$.
- **MOND:** In low-acceleration environments, environmentally active Temporal Shear supplies a candidate common origin for MOND-like scaling. This paper establishes the acceleration-scale correspondence but does not independently derive the complete MOND interpolation law.

TEP provides a single temporal-gradient framework within which CDM-like large-scale phenomenology and MOND-like low-acceleration behaviour may emerge in different environmental projections.

Box 6.2: Relation to Low-Acceleration Phenomenology

In TEP, MOND-like phenomenology is associated with the environmental activation of observable Temporal Shear in low-acceleration, weakly suppressed systems. The canonical screening operator $\mathcal{S}_\Sigma(\mathcal{E})$ becomes weak in low-gradient galactic environments, allowing the observable Temporal Shear response to become active. This supplies a candidate origin for MOND-like phenomenology without modifying the local value of c.

The empirical coincidence

$$a_T \sim cH_0$$

identifies the relevant observational acceleration scale, where H_0 is the conventional inverse-time scale encoded by the cosmological clock map rather than physical spatial expansion. In TEP, H_0 is not identified with $\dot{a}_m/a_m$, since $a_m = 1$.

Numerical Check:

$$a_0 \sim cH_0 \approx (3 \times 10^8\ \mathrm{m/s}) \times (2.2 \times 10^{-18}\ \mathrm{s}^{-1}) \approx 6.6 \times 10^{-10}\ \mathrm{m/s}^2$$

This lies within a factor of about five of the empirical MOND value $a_0 \approx 1.2 \times 10^{-10}$ m/s². The canonical TEP acceleration scale is the derived cosmological-horizon scale $g_t = cH_0/(2|\beta_A|) \approx 3.4 \times 10^{-10}$ m/s² (Paper 0 §2.2, v0.16); the SPARC-extracted empirical benchmark $g_{\mathrm{TEP}} \approx 5 \times 10^{-10}$ m/s² (Paper 0 §7; Paper 10) independently corroborates the same order and matches the ambient Galactic shear $a_{\mathrm{amb}} \approx 3.9 \times 10^{-10}$ m/s² at the solar circle. The environmental transition is supplied by the nested two-body operator $\mathcal{R}(s) = \mathcal{S}_\Sigma(X_{\mathrm{env}})^2\, y(s)$ — the effective pairwise projection of the master environmental operator $\mathcal{S}_\Sigma(\mathcal{E})$ under the selected two-branch kinetic completion $P(X,\phi) = -V(\phi) + \frac{2k}{3\Lambda_X^2} X|X|^{1/2} + X|X|/\Lambda_X^4$ ($P_{,X} = k\sqrt{\xi} + 2\xi$, $k = 16.03$; with $P = X - V + X|X|/\Lambda_X^4$ as the retained unit-term high-gradient benchmark; Paper 0 §2.2, v0.16, machine-checked benchmark table). TEP attributes this cross-scale relation to the same temporal-gradient structure represented by $\mathcal{S}_\Sigma(\mathcal{E})$. The defensible statement is that TEP produces an acceleration scale of cosmological magnitude, which suggests that TEP and MOND phenomenology may share a common origin. This matches the level of claim established in TEP-UCD (Paper 6); TEP-GL does not independently derive the numerical MOND acceleration scale.

Physical Interpretation: Low-acceleration phenomenology arises in TEP because the environmental operator $\mathcal{S}_\Sigma(\mathcal{E})$ leaves the Temporal Shear response active precisely in low-gradient regions. Regions with internal accelerations $a \lesssim g_t$ are weakly screened and the temporal gradient contributes; regions with $a \gtrsim g_t$ are strongly screened and follow Newtonian dynamics. The low-gradient scalar profile, its metric backreaction, and the reconstruction response must be fitted jointly to galaxy and lensing data before the MOND transition can be claimed as derived.

6.7 Convergence of Evidence

While TEP-GL is a theoretical proposal, it is notable that multiple independent anomalies in current cosmology converge on the phenomenology predicted by this framework. These tensions, often treated as separate puzzles, may represent a single systemic failure of the isochrony assumption.

TEP-GL Prediction	Existing Observational Anomaly	Status
Source-Dependent Shear (Kinematic noise bias; bounded at $\gamma_{\mathrm{TC}}^2/\sigma_{\mathrm{sh}}^2 \sim 4 \times 10^{-12}$, Box 5.3)	**S_8 Tension** Survey-dependent: DES Y6 lower than CMB; KiDS-Legacy consistent with Planck.	**Motivating Consistency**
Mass-Sheet Degeneracy (Temporal vs Spatial)	**H_0 Tension** Time-delay cosmography ($H_0 \approx 73$) conflicts with CMB ($H_0 \approx 67$). The chronometric entry is a degeneracy statement, not a supply channel: a common-mode endpoint rescaling acts on inferred $D_{\Delta t}$ exactly as an external convergence sheet does, but under the present-epoch convention its amplitude is the uniform local-well residual $A_o \approx 1 - 10^{-7}$ — far below the tension and identical for every lens system. The full channel audit on the real TDCOSMO-2025 systems (Paper 19, Step 58) confirms the propagated sector is clean at $\lesssim 10^{-8}$ fractional (~ 10 ms ceilings on 10 R_{E} traverses) and finds the only order-percent route to be the dynamical–lensing mass difference in the velocity-dispersion prior — in the direction that elevates inferred H_0 — while the mundane mass-model floor is $\sim 6\%$	**Degeneracy-level**

TEP-GL Prediction	Existing Observational Anomaly	Status
	(SDSS1206+4332 power-law vs composite). The corpus's quantitative H_0 channel is the Cepheid clock-bias sector (TEP-H0, Paper 11), a different projector.	
temporal-composite response (source-evolution response)	**Flux Ratio Anomalies** Substructure required to explain ratios is often not found; "phantom" substructure. TC contribution bounded at $\delta\mu/\mu \lesssim 10^{-5}$ (Box 5.2).	**Amplitude-bounded**
Variability-Shear Correlation	**Einstein Cross Anomalies** Existing Einstein Cross monitoring provides archival data suitable for testing the predicted variability-phase correlation; no dedicated TEP correlation analysis has yet been performed (Eigenbrod et al. 2008).	**Untested**

The theory supplies testable temporal channels on existing lensing and timing observations, with the source-independent optical convergence and Fermat delay evaluated directly from the unstretched-space temporal metric and two-branch action in §6.9 (Step 02) and Paper 19 (Step 64).

6.8 The Path Forward

The analysis shows that a new particulate dark component is not required by the temporal-field ontology: the same scalar configuration produces both the dynamical mass-inference response and the unstretched-space optical refraction (§6.9), alongside source-dependent reconstruction signatures. Full dark-sector closure is tested by two complementary observables — the matched dynamical–lensing phantom-mass ratio on isolated systems and the CDM-free absolute lensing potential across clusters and the cosmic web (§6.9; Paper 18 Step 21; Paper 19 Step 64). For particulate dark matter to remain necessary, these TEP predictions must be excluded at the required precision.

Three concrete consequences follow directly from the phantom-mass mechanism and require no further model-building:

Lensing radial-acceleration relation. On the historical isotropic-conformal benchmark ($\tilde{g}_{\mu\nu} = A^2 g_{\mu\nu}$), photon trajectories track the baryon-sourced geometry alone ($g_{\rm lens}^{\rm iso} \simeq g_{\rm bar}$), whereas under the canonical unstretched-space temporal metric ($\tilde{\gamma}_{ij} = \gamma_{ij}$, §6.9), intermediate temporal depth contributes half the dynamical scalar potential weight on single-pass rays ($g_{\rm lens}^{\rm u.s.} = g_{\rm bar} + \frac{1}{2}a_\phi$, giving $M_{\rm lens}^{\rm u.s.}/M_{\rm dyn} = (1+\beta_A^2 y_{\rm eff})/(1+2\beta_A^2 y_{\rm eff})$; Paper 0 Eq. 63, reaching $g_{\rm bar} + a_\phi$ on multi-component sightlines). In either case, the single-halo dynamical–lensing relation is predicted to be systematic in the radial-acceleration plane, not a scatter of independent anomalies, and is testable galaxy-by-galaxy wherever both inference channels exist.

Isolation dependence. The phantom mass is largest for isolated, diffuse, low-acceleration systems ($a \lesssim g_t$, $\mathcal{S}_\Sigma \to 1$) and is suppressed toward zero for galaxies embedded in high-acceleration environments — cluster members, compact groups, satellites in deep host potentials — where the environment screens the Temporal Shear. A dynamical mass excess that persists at full strength inside cluster cores would count directly against the mechanism.

Dissociative mergers. The Bullet Cluster is the discriminating case. The observed lensing peaks sit on the collisionless galaxies, offset from the dominant baryonic gas. Under isotropic conformal rescaling $\tilde{g}_{\mu\nu} = A^2(\phi) g_{\mu\nu}$, the conformal factor cancels out of null rays at every intermediate point, leaving only the spatial-gradient disformal channel $B(\partial\phi)^2$ (bounded $\sim$ 9–70$\times$ below the required convergence) and a compensated negative moat. Under the unstretched-space temporal metric (§6.9, Step 02), by contrast, intermediate clock rates $A(\phi(x))$ do not cancel against spatial stretching, and the selected two-branch action $P_{,X} = k\sqrt{\xi} + 2\xi$ reproduces the galaxy-peak convergence and offset morphology directly from the baryonic potential wells.

6.9 Explicit Ray Tracing on a Dissociative Geometry: Disformal Control vs. Unstretched-Space Temporal Refraction

The channel audit is now carried through explicitly at the level of the photon paths themselves (step_02). First, the disformal-only spatial-gradient route on the historical isotropic-conformal metric is evaluated as an adverse control, isolating why GW170817-bounded cone tilts cannot supply the cluster-scale convergence amplitude; second, the surviving cluster-lensing mechanism—direct optical refraction from the shared-lapse unstretched-space temporal metric—is evaluated on the identical four-component baryon-only geometry. In the isotropic-conformal plus disformal matter metric $\tilde{g}_{\mu\nu} = A^2(\phi) g_{\mu\nu} + B(\phi)\nabla_\mu\phi\nabla_\nu\phi$, the Sherman–Morrison inverse is $\tilde{g}^{\mu\nu} = A^{-2} g^{\mu\nu} - [B/(A^2(A^2+BQ))]\nabla^\mu\phi\nabla^\nu\phi$ with $Q = g^{\alpha\beta}\nabla_\alpha\phi\nabla_\beta\phi$, and null rays are the characteristics of $H = \frac{1}{2}\tilde{g}^{\mu\nu}k_\mu k_\nu = 0$: $\dot{x}^\mu = \tilde{g}^{\mu\nu}k_\nu$, $\dot{k}_\mu = -\frac{1}{2}\,\partial_\mu\tilde{g}^{\alpha\beta}k_\alpha k_\beta$. In the weak-field, perturbative-B limit the eikonal index is direction-dependent, $n_{\rm eff}(\hat{n}, \mathbf{x}) \simeq 1 - 2\Phi/c^2 + \frac{1}{2}(B/A^2)(\hat{n}\cdot\nabla\phi)^2$: photons are retarded only when propagating along the scalar gradient. The Born deflection $\alpha_i(\mathbf{b}) = \int \partial_i n_{\rm eff}\, dl$ and the reduced-deflection convergence $\kappa_{\rm dis}(\theta) = -\frac{1}{2}(D_{ls}/D_s)\, D_l\, \nabla_b \cdot \alpha$ were computed on a Bullet-like two-subcluster geometry (0.72 Mpc separation, gas centroid lagging the bullet galaxies by 0.24 Mpc, baryons only) with the screened field solved along each sightline, $\nabla u = y(g)\, g/c^2$. Three controls validate the implementation: the identical stencil applied to $n_{\rm GR} = -2\Phi/c^2$ reproduces the point-mass deflection $4GM/c^2 b$ to 1.6%; the full Hamiltonian system integrated with the exact $\tilde{g}^{\mu\nu}$ through a spherical scalar well, with the disformal-only deflection isolated by subtracting the $B_0 = 0$ ray, agrees with the Born reduction to 0.02–0.1 per cent at matched impact parameters once the on-ray field accumulation and the radial spline are truncated at the same radius (the earlier apparent shortfall was a mismatched accumulation domain, not an integration error); and the conformal sector contributes identically zero deflection.

The result answers the morphology half of the objection affirmatively. The disformal convergence concentrates on the collisionless scalar wells, not on the gas: at the bullet-galaxy sightline $\kappa_{\rm dis}/B_0 = 1.6 \times 10^{-7}$ versus 6.8×10^{-8} at the gas centroid, and the map peak sits on the main-cluster well to within a resolution cell. The kernel is compensated — a positive core on each well surrounded by a negative moat — which is itself a falsifiable structural signature distinguishing disformal-sourced from mass-sourced convergence. The "empty" lensing regions are not empty in the matter metric: they host the deepest ∇u.

The amplitude half remains adverse. Reproducing a peak convergence of $\kappa \simeq 0.3$ at the galaxy positions requires $B_0^{\rm req} \simeq 3.7 \times 10^5$ under the canonical halo convention — 4.8×10^3 above the GW170817-transferred bound $B_0 \lesssim 77.7$. Under the ambient-consistent convention (B reading the total field $u = \bar{u} + u_{\rm halo}$) the requirement drops to $B_0^{\rm req} \simeq 1.2 \times 10^{-3}$, but the same-convention multimessenger bound tightens to $B_0 \lesssim 3.1 \times 10^{-11}$ (Paper 19, step_63), worsening the exclusion to $\sim 4 \times 10^7$. Sourcing the field from the effective dynamical potential — the scalar wells tracking the phantom-enhanced $g_{\rm eff} = g(1 + 2\beta_A^2 y)$ rather than the bare baryonic field — deepens u and closes about 98% of the halo-convention gap ($B_0^{\rm req} \simeq 5.3 \times 10^3$, excluded by a factor ~ 68), but does not close it. The deficit is partially component-model dependent: stress-testing the Plummer scales (step_02b) shows that compactening all components improves the margin fastest, but only until the baryonic convergence alone overshoots the observed $\kappa \simeq 0.3$ at the galaxy sightline ($a \lesssim 0.85\, a_{\rm nom}$ is excluded by the lensing data itself); at the boundary $a \simeq 0.9\, a_{\rm nom}$ the requirement improves to $B_0^{\rm req} \simeq 720$, a factor ~ 9 over the bound. The residual gap therefore spans $\sim$ 9–70 across the defensible baryonic-model range, insensitive to the line-of-sight integration extent (doubling to ± 8 Mpc moves the margin by 13%). The corpus's admissible nested coupling was also evaluated on this geometry: with $B_{\rm eff} = B(\phi_{\rm tot})\, G(\delta u)$ — the envelope reading the total nested field while the gate $G = g_{\rm sat} \min(1, \delta u / u_{\rm act})$ closes on the smooth ambient and saturates inside nonlinear wells — and self-consistent sourcing, the requirement is $B_0^{\rm req} \simeq 0.02$–$0.15$ across the gate bracket $g_{\rm sat} = 10^{-3}$–7×10^{-3}, against the same-convention host-well bound $B_0 \lesssim 2.1 \times 10^{-5}$: an exclusion of order 10^3, consistent with the Paper-19 gate scan. The inference-side escape route is also bounded: the conformal-depth difference between the gas- and galaxy-peak sightlines is $\Delta u \simeq -3.0 \times 10^{-6}$ ($c\,\Delta u \simeq -0.9\ {\rm km\,s^{-1}}$), corresponding to an X-ray emissivity and gas-mass inference bias of order 10^{-6} — five orders of magnitude below what relocating the baryon dominance requires (and in any individual dissociative merger a $\sim 1\ {\rm km\,s^{-1}}$ differential clock shift is masked by $\mathcal{O}(10^2$--$10^3)\ {\rm km\,s^{-1}}$ hydrodynamic shock and ram-pressure stripping velocities, making it a baryon-census consistency bound and a regime accessible only in large ensemble-stacked cluster gravitational-redshift surveys).

The diagnosis above isolates why the spatial-gradient disformal channel $B(\phi)\partial_\mu\phi\partial_\nu\phi$ is suppressed: under isotropic conformal rescaling $\tilde{g}_{\mu\nu} = A^2(\phi) g_{\mu\nu}$, spatial rulers stretch by the same factor $A^2(\phi)$ as clocks, so $A^2(\phi(x))$ divides out of the null condition $d\tilde{s}^2 = 0$ at every intermediate point along the ray, killing the linear $\mathcal{O}(u) \sim 10^{-5}$ temporal depth and leaving only the quadratic $\mathcal{O}((\nabla u)^2)$ cone tilt. In TEP's cosmological ontology, however, space does not stretch: under the unstretched-space temporal metric $\tilde{g}_{\mu\nu} = -A_{\rm loc}^2(u_{\rm grav}(\mathbf{x}))U_\mu U_\nu + A_{\rm sp}^2(u_{\rm drift}(t,\mathbf{x}))\Pi_{\mu\nu}$ ($U_\mu \equiv -\nabla_\mu\bar{\phi}/\sqrt{2\bar{X}}$), the $(3+1)$ ADM spatial geometry $\gamma_{ij} = (1 - 2\Phi_N/c^2)\delta_{ij}$ remains unstretched while the spacetime lapse rescales as $N_{\rm TEP}(x) = A(\phi(x))N_{\rm GR}(x)$ along the temporal flow. Along a null ray across static space, intermediate clock rates $A(\phi(x))$ between source and observer no longer cancel against spatial stretching: $c\,dt/dl = A(\phi(x))^{-1}\sqrt{(1 - 2\Phi_N/c^2)/(1 + 2\Phi_N/c^2)} \simeq 1 - 2\Phi_N(x)/c^2 - \beta_A u_{\rm grav}(x)$, with transverse deflection gradient $-\beta_A \nabla_\perp u = \vec{a}_{\phi,\perp}/c^2$. Evaluated on the same four-component baryon-only Bullet geometry under the selected two-branch kinetic sector $P_{,X} = k\sqrt{\xi} + 2\xi$ ($k = 4\sqrt{2}\,|\beta_A| g_t/a_0 = 16.03$, fixed external calibration with zero per-system free parameters; `unstretched_temporal_depth_channel` in `step_02`), intermediate-point temporal refraction yields $\kappa_{\rm temp} = 0.065$ (1× temporal lapse) to 0.131 (2× dynamical normalization) on top of $\kappa_{\rm bar} = 0.241$ at the bullet-galaxy sightline, giving $\kappa_{\rm tot} = 0.307$–0.372 against the observed target $\kappa \simeq 0.30$. The bullet-half convergence peak lands at $x = +0.352$ Mpc (on the galaxies at $+0.36$ Mpc, versus $\kappa_{\rm tot} = 0.223$–0.284 at the shocked-gas centroid at $+0.12$ Mpc), $\kappa_{\rm temp}$ is positive everywhere across the field ($\min \kappa_{\rm temp} = +4.6 \times 10^{-3}$, eliminating the negative moat), Solar-System Shapiro bounds remain satisfied via the quadratic-branch screening $y \sim (r/r_*)^{4/3} \ll 10^{-5}$, and $c_{\rm GW} = c_\gamma$ holds identically. Specifically, although $-(A^2 - 1)U_\mu U_\nu = \frac{1 - A^2}{2\bar{X}}\nabla_\mu\bar{\phi}\nabla_\nu\bar{\phi}$ has the algebraic form of a longitudinal disformal shift along the cosmic temporal flow, it is not a photon-only disformal coupling on an un-lapsed tensor metric (which would split photon and gravitational-wave transit times by $\Delta t \sim \int |\beta_A| u\, dl/c \sim 10^7$ s across Galactic and host haloes, the dark-matter-emulator exclusion of Boran et al. 2018, and invert the longitudinal sign in voids where $A > 1$); because proper time $d\tilde{t} = A_{\rm loc}(u(\mathbf{x}))\, dt$ is the universal physical time of the inhomogeneous ADM foliation itself, varying the spatially covariant ADM/DHOST gravitational action $S_{\rm grav} = \frac{1}{2}M_{\rm Pl}^2 \int d^3x\, dt\, \sqrt{\gamma}\, N_{\rm GR}[R^{(3)}(\gamma) + \tilde{K}_{ij}\tilde{K}^{ij} - \tilde{K}^2]$ ($\tilde{K}_{ij} = (2N_{\rm TEP}c)^{-1}\partial_t\gamma_{ij} = A_{\rm loc}(u)^{-1}K_{ij}$, with $\delta S_{\rm grav}/\delta u \equiv 0$ in the static limit $K_{ij} \to 0$ preserving the Poisson and scalar source equations without Jordan-frame degeneracy; Paper 0, §2.2, Step 80) with respect to transverse-traceless tensor perturbations $h_{ij}^{\rm TT}$ of γ_{ij} and varying $S_{\rm EM}[\tilde{g}]$ with respect to electromagnetic waves $A_i^{\rm T}$ yields the identical quadratic wave operator $\Box_{\rm TEP} = -(N_{\rm TEP}c)^{-2}\partial_t^2 + \gamma^{ij}\partial_i\partial_j$ on the unstretched spatial metric γ_{ij}, experiencing the identical refractive index $n(x)$ across wells and voids alike ($\Delta t_{\gamma-{\rm GW}} = 0$ identically).

The Bullet Cluster therefore sharply discriminates between the two metric realizations: under isotropic conformal stretching the quadratic $B(\partial\phi)^2$ channel falls short of current bounds by a factor $\sim$ 9–70 and produces a compensated negative moat, whereas under the unstretched-space temporal metric the selected two-branch action reproduces the $\kappa \simeq 0.30$ galaxy-peak convergence and offset morphology from baryons and intermediate temporal depth alone, with the differential conformal clock bias on the X-ray gas census bounded to the 10^{-6} level.

7. Conclusions

7.1 Limitations of the Isochrony Axiom

For nearly a century, dark-matter inference has been developed within a framework that treats the residual source-time structure relevant to static reconstruction as effectively synchronous once standard propagation delays are modeled. By treating the photons in telescopes as representing a single spatial slice of the source, standard models have been forced to interpret all observed distortions as spatial deflections caused by mass. It has been shown that this *Isochrony Axiom* is an effective approximation that may break down in the presence of generalized metric couplings. In the TEP ontology, the dark sector is not fundamental matter or energy; it is part of the residual produced when dynamical temporal transport is forced into a synchronous spatial reconstruction. The same interpretive principle applies cosmologically: observed redshift is reconstructed conventionally as physical spatial expansion, whereas TEP attributes it to the conformal proper-time map on an underlying static spatial geometry (Athens, Paper 26; Thika, Paper 27). Tortola develops the corresponding lensing-sector result: temporal-field structure can be reconstructed as spatial mass.

Under isotropic conformal rescaling $\tilde{g}_{\mu\nu} = A^2(\phi) g_{\mu\nu}$, spatial rulers stretch by $A^2(\phi)$ alongside clocks so $A^2(\phi(x))$ cancels out of $d\tilde{s}^2 = 0$, leaving only the Planck-suppressed scalar stress tensor ($\kappa_\phi \simeq 6.5 \times 10^{-6}$ on the selected sector; Paper 19, Step 64) and the bounded disformal cone tilt (Step 63). Under the unstretched-space temporal metric $\tilde{g}_{\mu\nu} = -A^2_{\rm loc}(u_{\rm grav}(\mathbf{x}))U_\mu U_\nu + A^2_{\rm sp}(u_{\rm drift}(t,\mathbf{x}))\Pi_{\mu\nu}$ ($U_\mu \equiv -\nabla_\mu\bar{\phi}/\sqrt{2\bar{X}}$), space is static and does not stretch, so intermediate clock rates $A(\phi(x))$ along the photon path do not cancel: every intermediate point x contributes its local temporal depth $-\beta_A u_{\rm grav}(x)$ directly to the optical refractive index $n(x) \simeq 1 - 2\Phi_N(x)/c^2 - \beta_A u_{\rm grav}(x)$, reproducing the Bullet Cluster galaxy-peak convergence ($\kappa_{\rm tot} = 0.307$–0.372, no negative moat; Step 02), the SN Refsdal Fermat delay and arrival order (Paper 19, Step 64), and the Planck CMB lensing cosmic-web boost (Paper 18, Step 21) from baryons and the selected two-branch action $P_{,X} = k\sqrt{\xi} + 2\xi$ with zero particulate dark matter.

7.2 Summary of Findings

1. **Unified Dynamical and Optical Refraction from Unstretched Space:** While isotropic conformal rescaling cancels out of null geodesics and leaves only $\mathcal{O}(u^2)$ scalar stress-energy, TEP's unstretched-space temporal metric retains the linear $\mathcal{O}(u)$ intermediate temporal depth along every photon path. Sourced by baryons on the zero-free-parameter selected two-branch action $P_{,X} = k\sqrt{\xi} + 2\xi$, it supplies both the dynamical Temporal Shear on slow matter and the source-independent optical convergence across galaxies, clusters (Step 02), strong-lensing delays (Paper 19, Step 64), and CMB lensing (Paper 18, Step 21), alongside the bounded source-dependent temporal-composite response for evolving sources.
2. **The GW170817 "Speed Limit" and Solar-System Screening:** While GW170817 constrains disformal cone-tilt propagation speeds to $|c_\gamma - c_g|/c \lesssim 10^{-15}$, the unstretched-space temporal metric preserves $c_{\rm GW} = c_\gamma$ identically because both tensor and electromagnetic wave sectors evolve in the same physical ADM lapse $d\tilde{t} = A(\phi)\,dt$ on the unstretched spatial manifold ($\Delta t_{\gamma-{\rm GW}} = 0$), and it satisfies Solar-System PPN Shapiro bounds via the quadratic-branch kinetic screening $y \sim (r/r_*)^{4/3} \ll 10^{-5}$ (Paper 0, Step 57).
3. **A New Observational Era:** Two regimes for testing the time-domain and optical channels are defined: millisecond-scale multipath bounds on the disformal sector in high-time-resolution astrophysics (FRBs, pulsars), variability-conditioned shear residuals testing the bounded temporal-composite envelope, and direct cluster, strong-lens, and ensemble-stacked cluster gravitational-redshift tests of the unstretched-space temporal convergence and conformal potential depth (Step 02).

7.3 Future Outlook: Time, Not Mass

The persistence of the dark matter problem despite decades of particle searches suggests a potential error in the underlying premises. The error may lie in the treatment of time. By treating time as a passive parameter rather than an active dynamical field, there is a risk of blinding analysis to the true nature of the "dark" universe.

The path forward lies in *Chronometric Lensing*. The field must move beyond static mass-mapping and towards *chronometric mapping*—measuring the detailed arrival-time structure of the universe. The dark-sector tests are the dynamical–lensing mass difference on matched low-acceleration systems and the absolute CDM-free lensing potential, including $C_\ell^{\phi\phi}$. Lensed-FRB residuals separately test the disformal transport sector. The focus of inquiry must shift from searching for missing mass to characterizing the dynamical structure of time.

7.4 The Paradigm Shift

Within the operational axioms adopted here, the dynamical and optical phenomenology traditionally attributed to a particulate dark sector is reinterpreted as the projection of two-metric time transport onto inference pipelines that assume isochrony: Temporal Shear governs slow matter trajectories, while intermediate temporal depth across unstretched space refracts photons (§6.9, Step 02; Papers 18, 19), predicting both the baryon-sourced static lensing convergence and a testable single-halo dynamical–lensing mass ratio $M^{\rm u.s.}_{\rm lens}/M_{\rm dyn} = (1+\beta_A^2 y_{\rm eff})/(1+2\beta_A^2 y_{\rm eff})$. The anomalies are real; the claim is that their standard "missing mass" interpretation is not unique once the Isochrony Axiom is relaxed.

In dynamics, the dark sector is thus reinterpreted not as an invisible substance, but as the shadow of temporal transport on matter trajectories.

The observational program outlined herein—time-domain lensing of fast transients, precision strong-lens residual timing, source-variability-dependent shear consistency tests, and above all the environment-resolved dynamical–lensing mass measurement—provides a viable route to distinguish particle dark matter from temporal transport while maintaining a conservative baseline anchored to existing multi-messenger constraints, thereby offering a decisive, falsifiable avenue for confronting the dark matter problem.

Updated Operational Interpretation

The later dedicated strong-lensing analysis (TEP-LENS, Paper 19) refines the observational implementation used here. In particular, ordinary image-arrival delays remain algebraically integrable; the strong-lensing test is formulated as an observed-versus-GR-predicted transport residual rather than a literal closure violation. The framework presented here is consistent with that refinement: Axiom 3 already distinguishes open-path differential residuals (the GL observable) from closed-loop holonomy, and on the historical isotropic-conformal benchmark ($\tilde{g}_{\mu\nu} = A^2 g_{\mu\nu}$) the static clock-transfer contribution is identified as a clock-transfer discrepancy rather than a direct refraction of null geodesics. The two papers state one unified answer across both metric realizations: on the isotropic-conformal benchmark, the exact additional static-conformal inter-image residual is 0.0 d (Paper 19, step_56), and every year-scale figure quoted here is a reconstruction-space amplitude with the spatial-disformal differential channel bounded at the ms–s level; under the canonical unified unstretched-space temporal metric ($\tilde{\gamma}_{ij} = \gamma_{ij}$ at fixed cosmic epoch; Paper 0 §2.2, §6.9 Step 02, and Paper 19 Step 64), intermediate temporal depth $-\beta_A u_{\rm grav}(\mathbf{x})$ on the selected two-branch sector $P_{,X} = k\sqrt{\xi} + 2\xi$ ($k = 16.03$) directly refracts photons along the line of sight, reproducing both the Bullet Cluster galaxy-peak convergence ($\kappa_{\rm tot} = 0.307$--0.372) and the SN Refsdal Fermat delay ($+330.9$ d sheet and $+221.6$--443.2 d 3-D NFW ray) from $f_b = 0.16$ baryons alone. Readers should consult TEP-LENS for the current canonical strong-lensing time-delay formulation.

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Data Availability & Reproducibility

This work follows open-science practices. This is a theoretical framework paper developing the TEP-GL (Temporal Equivalence Principle - Gravitational Lensing) model. The analysis builds upon established dark matter phenomenology from the literature and derives new observational predictions from first principles.

Repository & Code

GitHub Repository: github.com/matthewsmawfield/TEP-GL

The repository contains the complete theoretical framework, mathematical derivations, and observational predictions for the TEP-GL model.

Repository Structure

```
TEP-GL/
├── manuscripts/               # Markdown manuscript sources
│   └── manuscript-tep-gl.md
├── site/
│   ├── components/            # HTML manuscript sections
│   │   ├── abstract.html
│   │   ├── section_1_introduction.html
│   │   ├── section_2_theoretical_framework.html
│   │   ├── section_3_quantitative_comparison.html
│   │   ├── section_4_gw170817.html
│   │   ├── section_5_observational_predictions.html
│   │   ├── section_6_discussion.html
│   │   ├── section_7_conclusions.html
│   │   ├── acknowledgments.html
│   │   ├── references.html
│   │   └── appendix_a.html
│   └── manifest.json
├── scripts/                   # Analysis scripts
│   ├── steps/
│   │   └── step_01_amplitude_ledger.py   # temporal-composite amplitude
│   │                                     # ledger (Boxes 3.2, 3.3, 5.1–5.3)
│   └── utils/
├── results/
│   └── step_01_amplitude_ledger.json    # machine-readable ledger output
├── requirements.txt           # Python dependencies
├── CITATION.cff               # Citation metadata
└── README.md                  # Repository documentation
```

The amplitude bookkeeping quoted in Boxes 3.2, 3.3 and 5.1–5.3 — the source proper motion, the reconstruction-space and propagating-carrier temporal-composite amplitudes, the standard delay-map baseline $\gamma^{\rm std} = \mu_s\, \partial_\theta \Delta T_{\rm Fermat}$ of §3.4/§5.2, the delay-field amplitude probed by each falsification threshold, and the exact-one-form cancellation audit of §3.1.3 (the endpoint-determination of the shear integral on two multipath rays of the Box-3.1 toy halo) — is reproduced by `scripts/steps/step_01_amplitude_ledger.py`, whose output is archived at `results/step_01_amplitude_ledger.json`.

Data Provenance

This is a theoretical framework paper. Primary data references:

- **GW170817:** LIGO/Virgo gravitational wave and EM counterpart data (public)
- **Dark matter phenomenology:** Established lensing results from the literature (fully cited)
- **SPARC database:** For TEP-UCD cross-paper validation

Reproduction Instructions

Quick Start (Manuscript Build)

```
# 1. Clone repository
git clone https://github.com/matthewsmawfield/TEP-GL.git
cd TEP-GL

# 2. Build manuscript website
cd site
npm install
npm run build

# 3. Output in site/dist/
```

System Requirements

- **Node.js** 16+ (for site building)
- **Storage:** < 50 MB

Theoretical Framework Documentation

The TEP-GL framework derives from three foundational postulates:

1. **Two-Metric Postulate:** Gravitational dynamics and universal causal matter coupling are distinguished by $g_{\mu\nu}$ and $\tilde{g}_{\mu\nu}$.
2. **Temporal Transport:** Conformal structure produces open-path clock/frequency transport, while residual closed-loop synchronization non-integrability requires a disformal or otherwise non-exact sector.
3. **Chronometric Reconstruction:** Temporal-field transport and scalar-induced geometry can project into conventionally inferred spatial mass.

Software Versions

- **Node.js** 16+
- **Python** 3.8+ (optional utilities)

Appendix A: Mathematical Derivations

A.1 Proof of Null Cone Invariance in the Conformal Limit

Proposition: If two metrics are conformally related by $\tilde{g}_{\mu\nu} = A^2(\phi) g_{\mu\nu}$, they share the same null geodesics as unparameterized curves.

Proof:
Let $k^\mu = dx^\mu/d\lambda$ be a null vector in $g_{\mu\nu}$, satisfying $g_{\mu\nu}k^\mu k^\nu = 0$ and the geodesic equation $k^\nu \nabla_\nu k^\mu = 0$ (where ∇ is the Levi-Civita connection of g).
In the metric $\tilde{g}_{\mu\nu}$, the null condition holds immediately:

$$\tilde{g}_{\mu\nu}k^\mu k^\nu = A^2(\phi) g_{\mu\nu}k^\mu k^\nu = 0$$

The connection coefficients $\tilde{\Gamma}^\lambda_{\mu\nu}$ for $\tilde{g}$ are related to $\Gamma^\lambda_{\mu\nu}$ by:

$$\tilde{\Gamma}^\lambda_{\mu\nu} = \Gamma^\lambda_{\mu\nu} + \delta^\lambda_\mu \partial_\nu \ln A(\phi) + \delta^\lambda_\nu \partial_\mu \ln A(\phi) - g_{\mu\nu} g^{\lambda\sigma} \partial_\sigma \ln A(\phi)$$

Substituting this into the geodesic equation for $\tilde{g}$:

$$k^\nu \tilde{\nabla}_\nu k^\mu = k^\nu \nabla_\nu k^\mu + 2k^\mu k^\nu \partial_\nu \ln A(\phi) - g_{\nu\sigma}k^\nu k^\sigma \dots$$

Since k is null ($g_{\nu\sigma}k^\nu k^\sigma = 0$) and $k^\nu \nabla_\nu k^\mu = 0$, this yields:

$$k^\nu \tilde{\nabla}_\nu k^\mu = 2(k^\nu \partial_\nu \ln A(\phi)) k^\mu$$

This is the geodesic equation with a non-affine parameterization ($Dk/d\lambda \propto k$). Thus, the curve is a geodesic of $\tilde{g}$, differing only by the parameterization (the clock rate). □